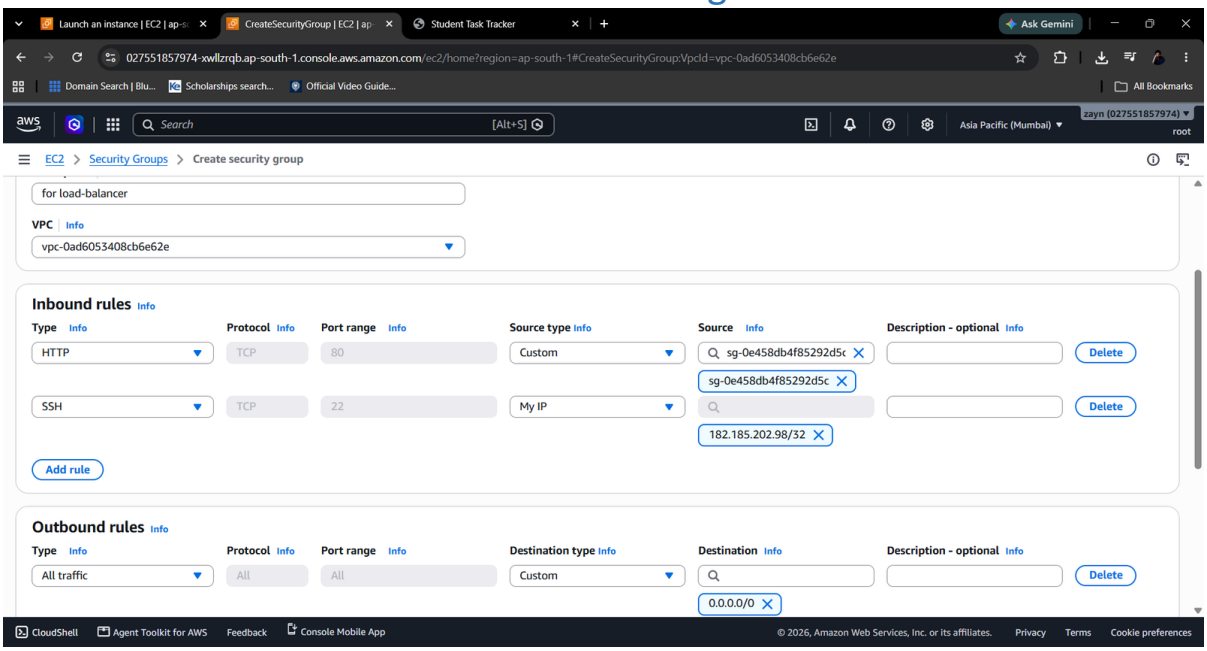
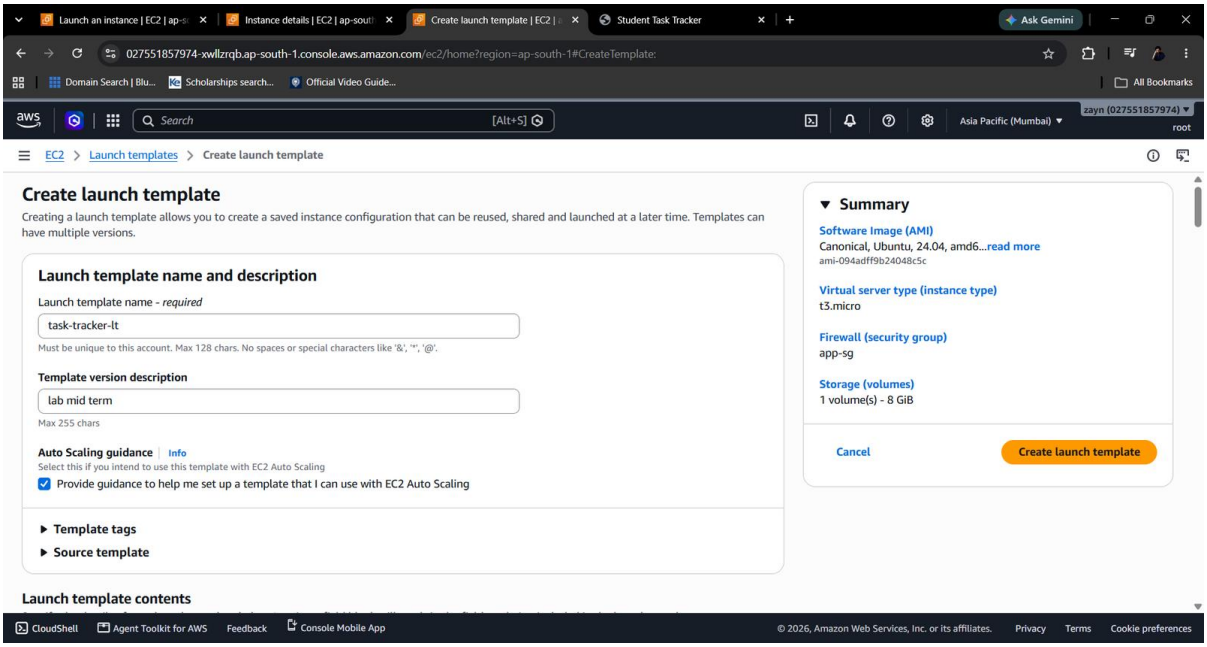
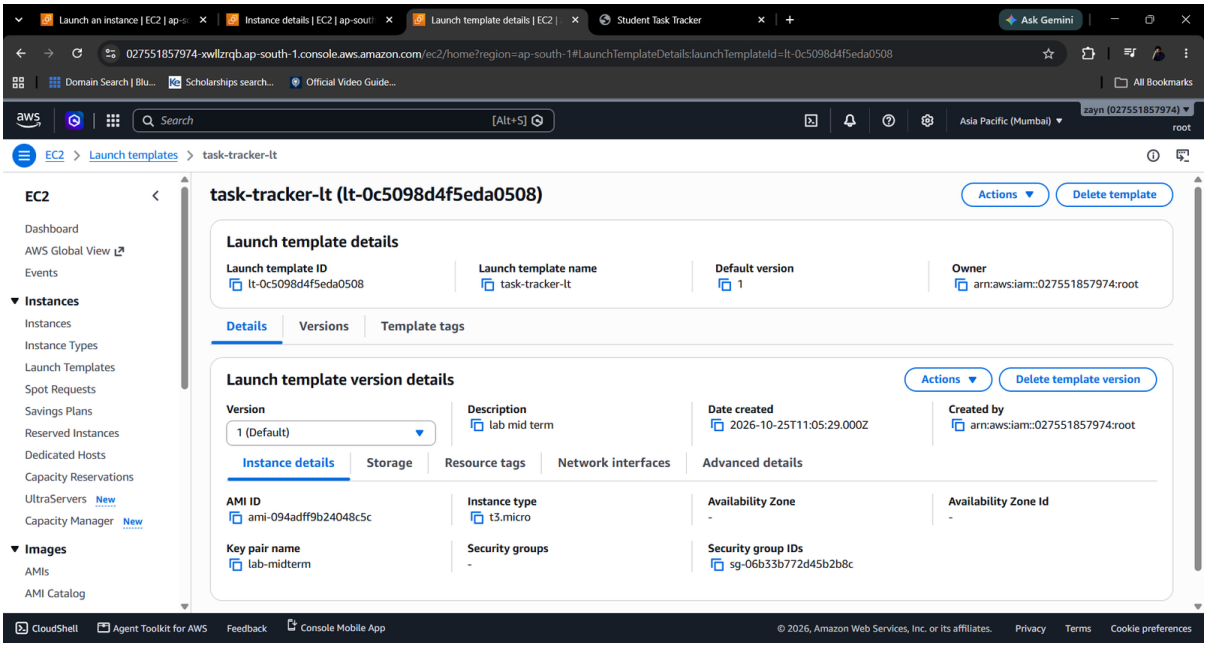
Name: Muhammad Zain Abbas
Roll No: SP23-BSE-136

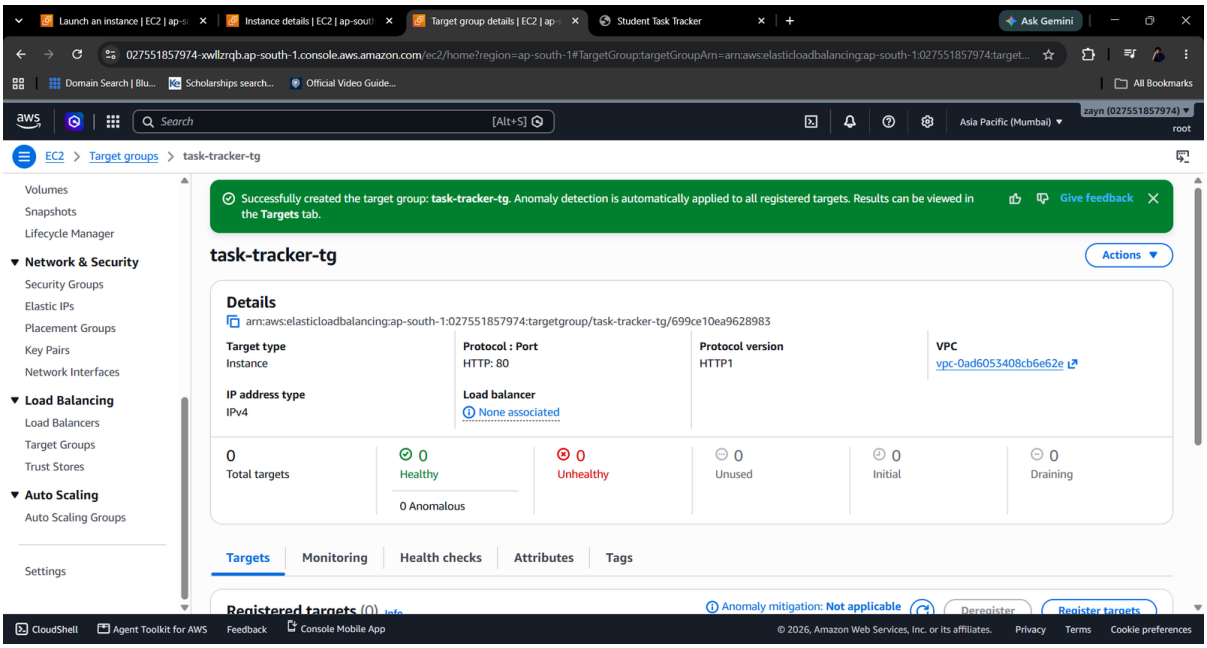
URL: http://task-tracker-alb-1135327115.ap-south-1.elb.amazonaws.com/

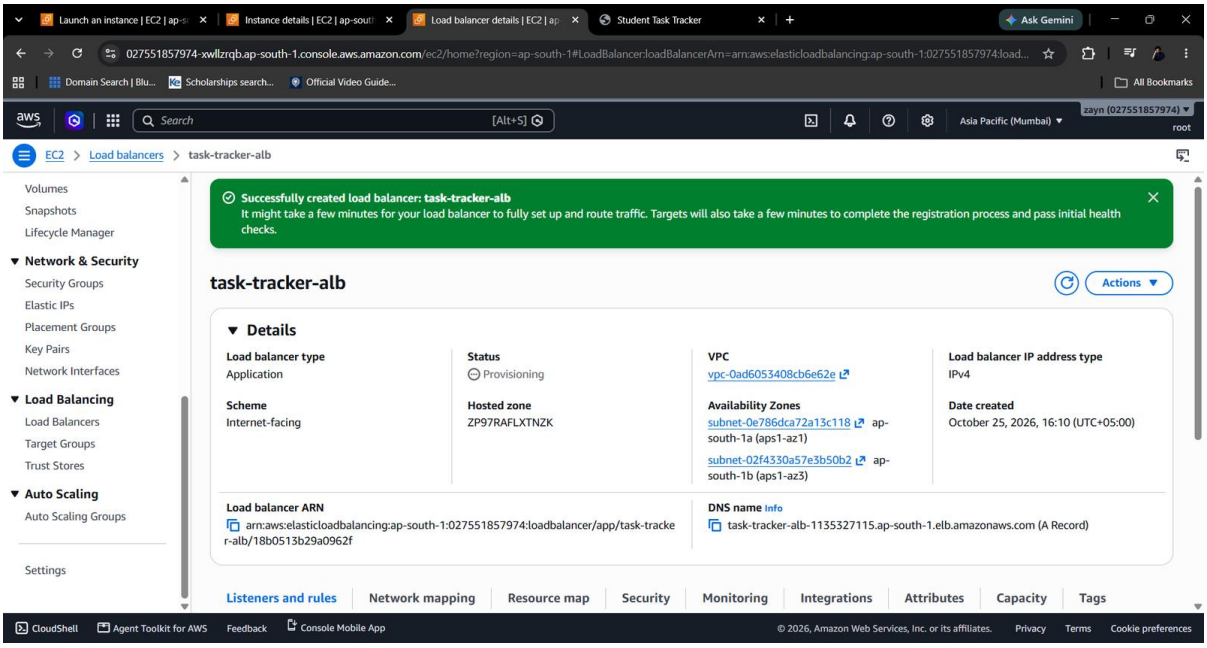
Task 1: AWS Load Balancer & Auto Scaling

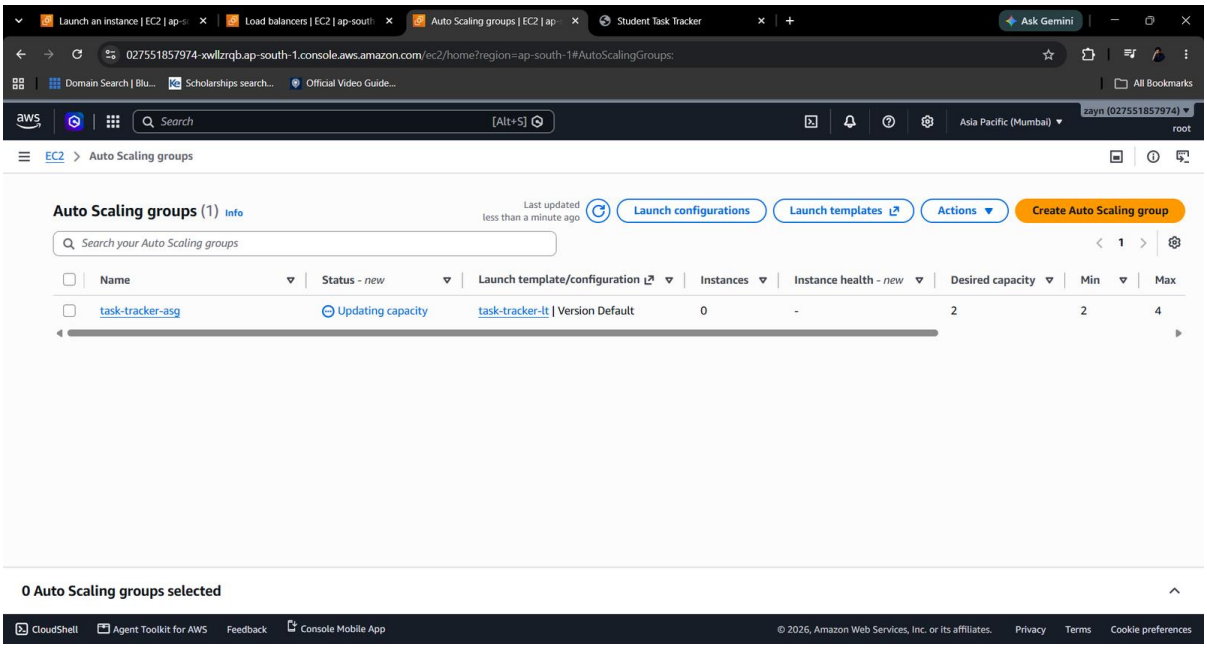


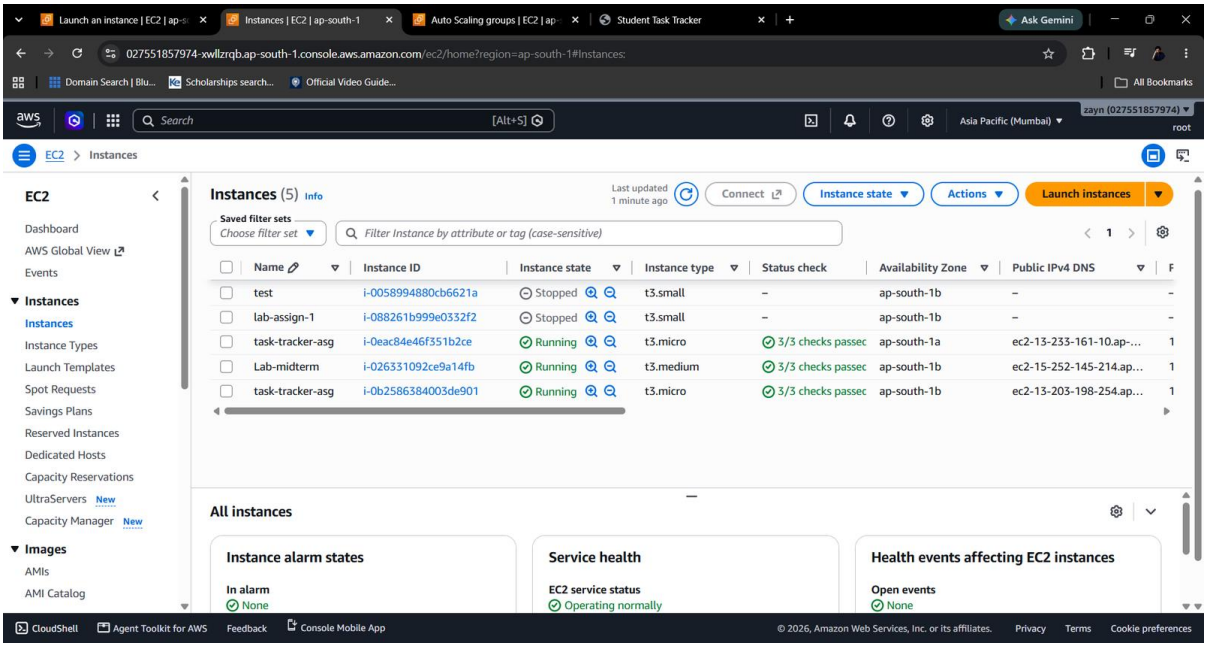


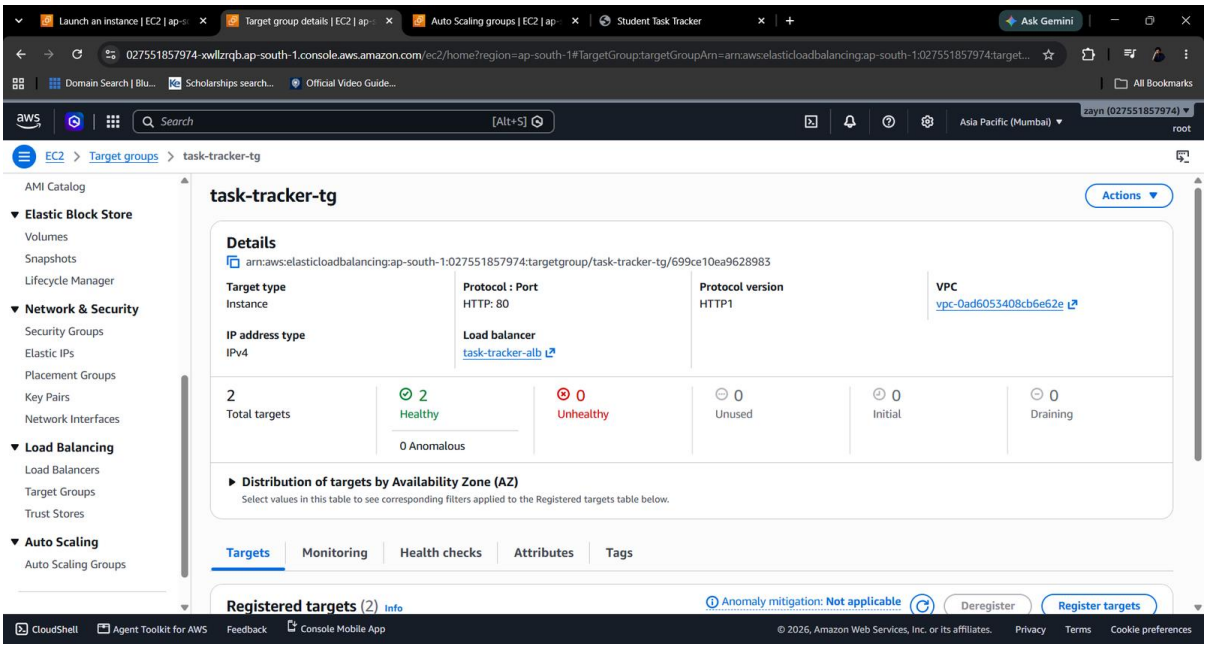

Launch an instance | EC2 | ap-s
Instance details | EC2 | ap-south
Launch template details | EC2 |
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchTemplateDetails:launchTemplateId=lt-0c5098d4f5eda0508
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Launch templates > task-tracker-lt
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
task-tracker-lt (lt-0c5098d4f5eda0508)
Actions
Delete template
Launch template details
Launch template ID
lt-0c5098d4f5eda0508
Launch template name
task-tracker-lt
Default version
1
Owner
arn:aws:iam::027551857974:root
Details
Versions
Template tags
Launch template version details
Actions
Delete template version
Version
1 (Default)
Description
lab mid term
Date created
2026-10-25T11:05:29.000Z
Created by
arn:aws:iam::027551857974:root
Instance details
Storage
Resource tags
Network interfaces
Advanced details
AMI ID
ami-094adff9b24048c5c
Instance type
t3.micro
Availability Zone
-
Availability Zone Id
-
Key pair name
lab-midterm
Security groups
-
Security group IDs
sg-06b33b772d45b2b8c
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Agent Toolkit for AWS
Feedback
Console Mobile App
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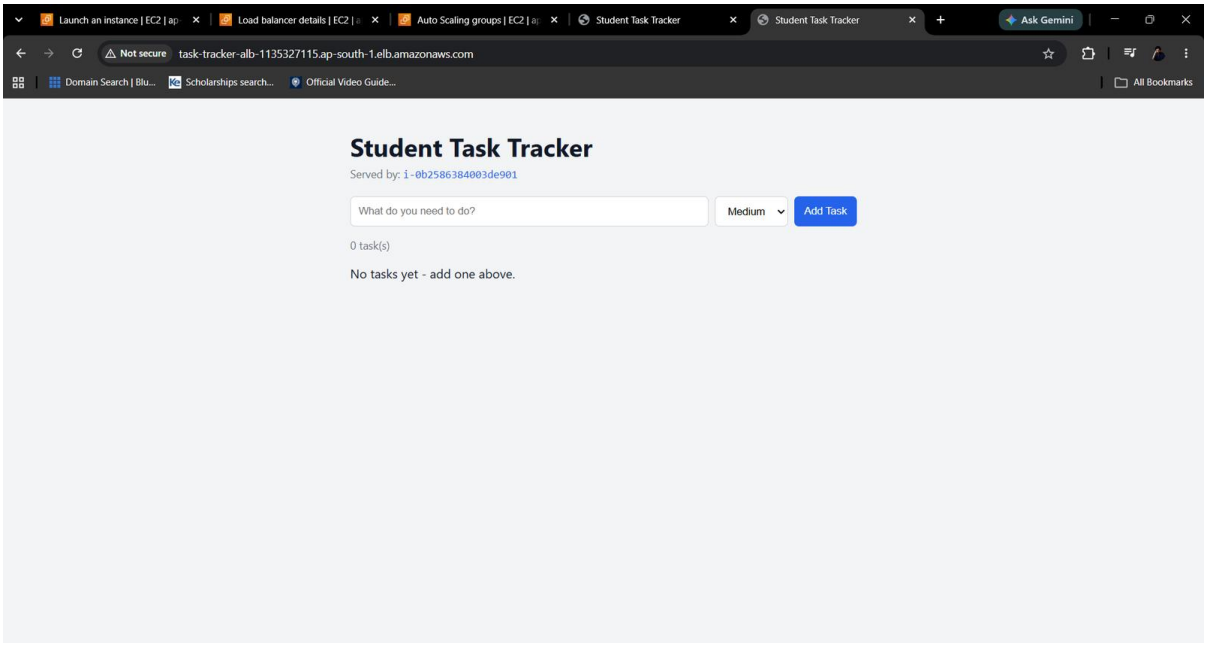

Launch an instance | EC2 | ap-s
Instance details | EC2 | ap-south
Target group details | EC2 | ap-
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroup:targetGroupArn=arn:aws:elasticloadbalancing:ap-south-1:027551857974:target...
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Target groups > task-tracker-tg
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Successfully created the target group: task-tracker-tg. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.
Give feedback
task-tracker-tg
Actions
Details
arn:aws:elasticloadbalancing:ap-south-1:027551857974:targetgroup/task-tracker-tg/699ce10ea9628983
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-0ad6053408cb6e62e
IP address type
IPv4
Load balancer
None associated
0
Total targets
0
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
Targets
Monitoring
Health checks
Attributes
Tags
Registered targets (0)
Anomaly mitigation: Not applicable
Deregister
Register targets
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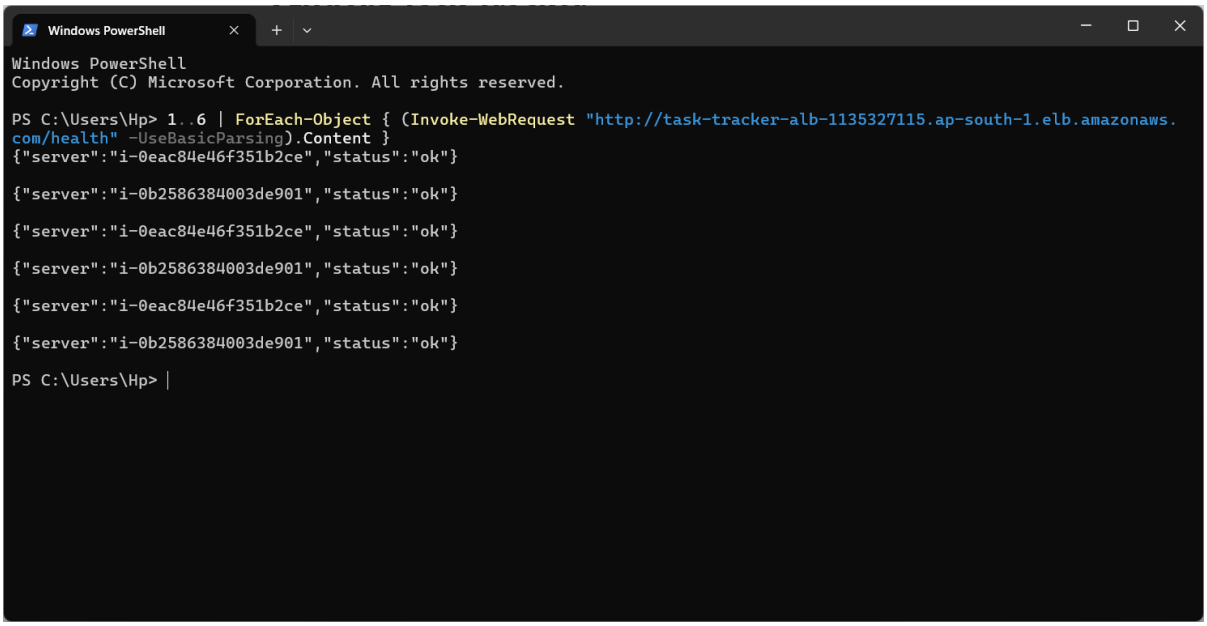

Launch an instance | EC2 | ap-s
Instance details | EC2 | ap-south
Load balancer details | EC2 | ap
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:ap-south-1:027551857974:load...
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Load balancers > task-tracker-alb
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Successfully created load balancer: task-tracker-alb
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.
task-tracker-alb
Actions
Details
Load balancer type
Application
Status
Provisioning
VPC
vpc-0ad6053408cb6e62e
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
ZP97RAFLXTNZK
Availability Zones
subnet-0e786dca72a13c118 ap-south-1a (aps1-az1)
subnet-02f4330a57e3b50b2 ap-south-1b (aps1-az3)
Date created
October 25, 2026, 16:10 (UTC+05:00)
Load balancer ARN
arn:aws:elasticloadbalancing:ap-south-1:027551857974:loadbalancer/app/task-tracker-alb/18b0513b29a0962f
DNS name Info
task-tracker-alb-1135327115.ap-south-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Attributes
Capacity
Tags
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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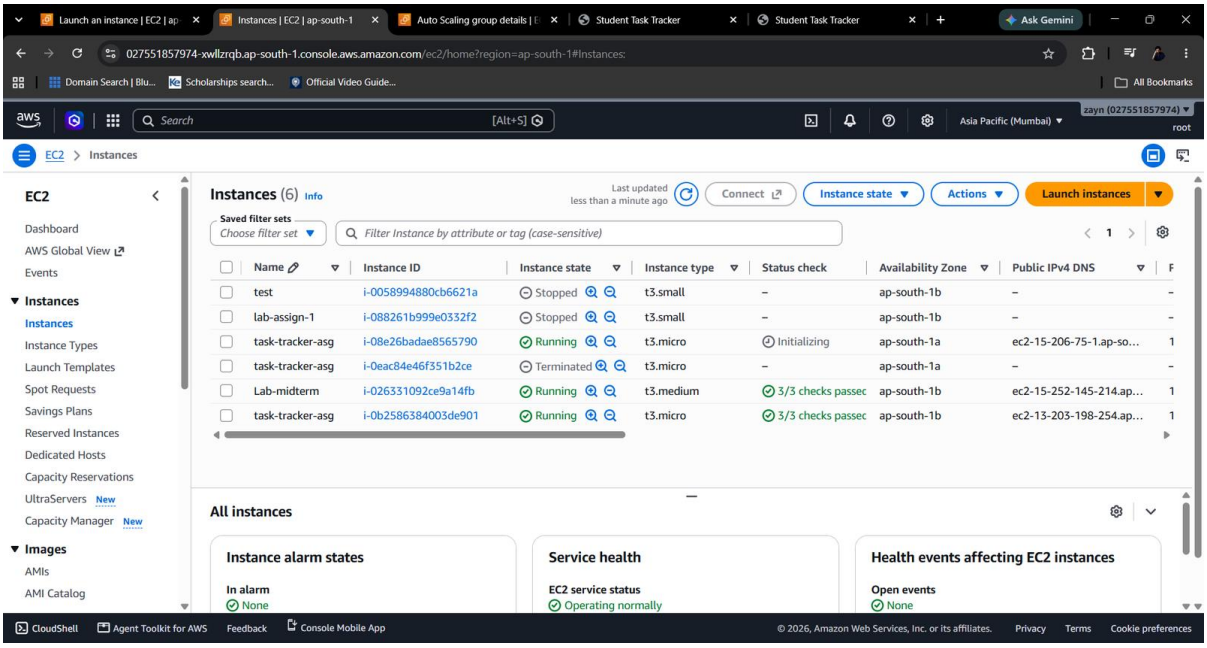

Launch an instance | EC2 | ap-s
Load balancers | EC2 | ap-south
Auto Scaling groups | EC2 | ap-
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AutoScalingGroups:
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Auto Scaling groups
Auto Scaling groups (1) Info
Last updated less than a minute ago
Launch configurations
Launch templates
Actions
Create Auto Scaling group
Search your Auto Scaling groups
1
Name
Status - new
Launch template/configuration
Instances
Instance health - new
Desired capacity
Min
Max
task-tracker-asg
Updating capacity
task-tracker-lt | Version Default
0
-
2
2
4
0 Auto Scaling groups selected
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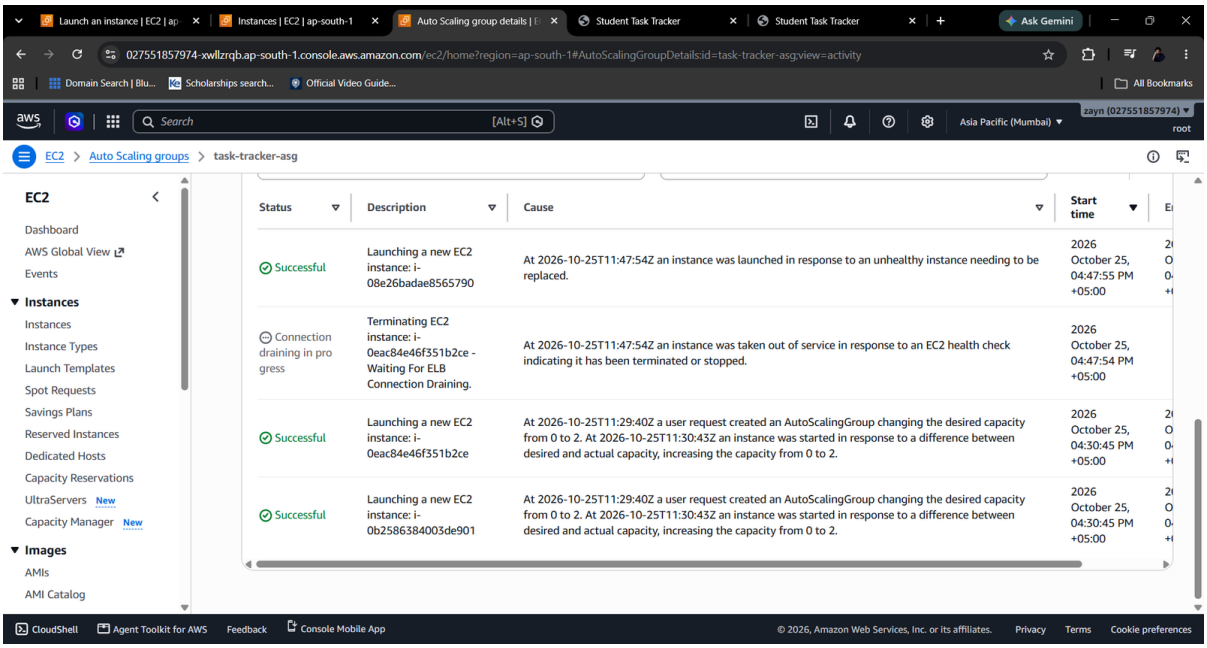

Launch an instance | EC2 | ap-s
Instances | EC2 | ap-south-1
Auto Scaling groups | EC2 | ap-
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Instances (5) Info
Last updated 1 minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
test i-0058994880cb6621a Stopped t3.small – ap-south-1b –
lab-assign-1 i-088261b999e0332f2 Stopped t3.small – ap-south-1b –
task-tracker-asg i-0eac84e46f351b2ce Running t3.micro 3/3 checks passed ap-south-1a ec2-13-233-161-10.ap-...
Lab-midterm i-026331092ce9a14fb Running t3.medium 3/3 checks passed ap-south-1b ec2-15-252-145-214.ap...
task-tracker-asg i-0b2586384003de901 Running t3.micro 3/3 checks passed ap-south-1b ec2-13-203-198-254.ap...
All instances
Instance alarm states
In alarm
None
Service health
EC2 service status
Operating normally
Health events affecting EC2 instances
Open events
None
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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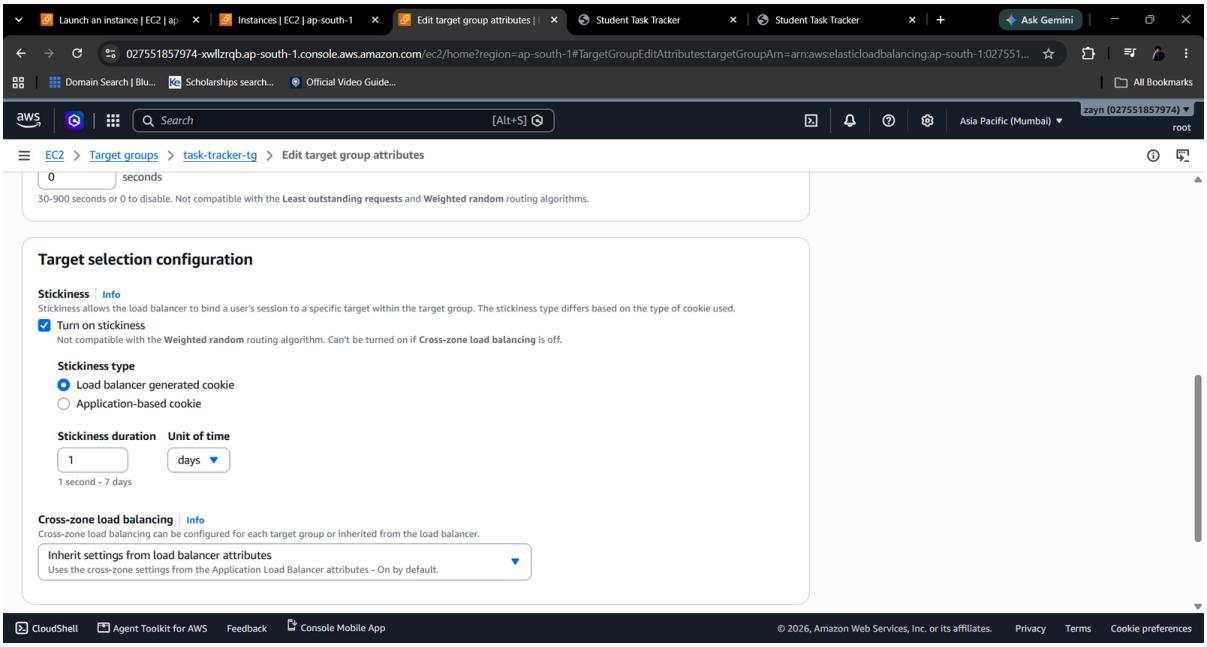

Launch an instance | EC2 | ap-s
Target group details | EC2 | ap-
Auto Scaling groups | EC2 | ap-
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroup:targetGroupArn=arn:aws:elasticloadbalancing:ap-south-1:027551857974:target...
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Target groups > task-tracker-tg
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
task-tracker-tg
Actions
Details
arn:aws:elasticloadbalancing:ap-south-1:027551857974:targetgroup/task-tracker-tg/699ce10ea9628983
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-0ad6053408cb6e62e
IP address type
IPv4
Load balancer
task-tracker-alb
2
Total targets
2
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.
Targets
Monitoring
Health checks
Attributes
Tags
Registered targets (2) Info
Anomaly mitigation: Not applicable
Deregister
Register targets
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Launch an instance | EC2 | ap
Load balancer details | EC2 |
Auto Scaling groups | EC2 | a
Student Task Tracker
Student Task Tracker
Ask Gemini
Not secure
task-tracker-alb-1135327115.ap-south-1.elb.amazonaws.com
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Student Task Tracker
Served by: i-0b2586384003de901
What do you need to do?
Medium
Add Task
0 task(s)
No tasks yet - add one above.

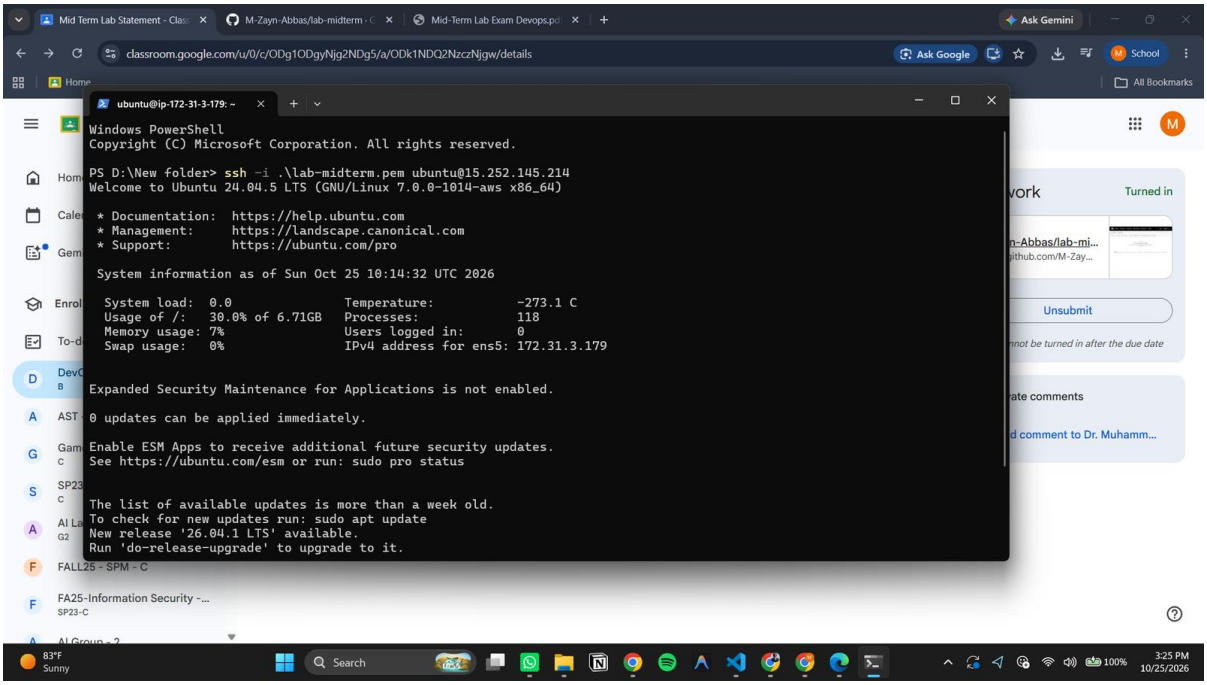

Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\Hp> 1..6 | ForEach-Object { (Invoke-WebRequest "http://task-tracker-alb-1135327115.ap-south-1.elb.amazonaws.com/health" -UseBasicParsing).Content }
{"server":"i-0eac84e46f351b2ce","status":"ok"}
{"server":"i-0b2586384003de901","status":"ok"}
{"server":"i-0eac84e46f351b2ce","status":"ok"}
{"server":"i-0b2586384003de901","status":"ok"}
{"server":"i-0eac84e46f351b2ce","status":"ok"}
{"server":"i-0b2586384003de901","status":"ok"}
PS C:\Users\Hp>

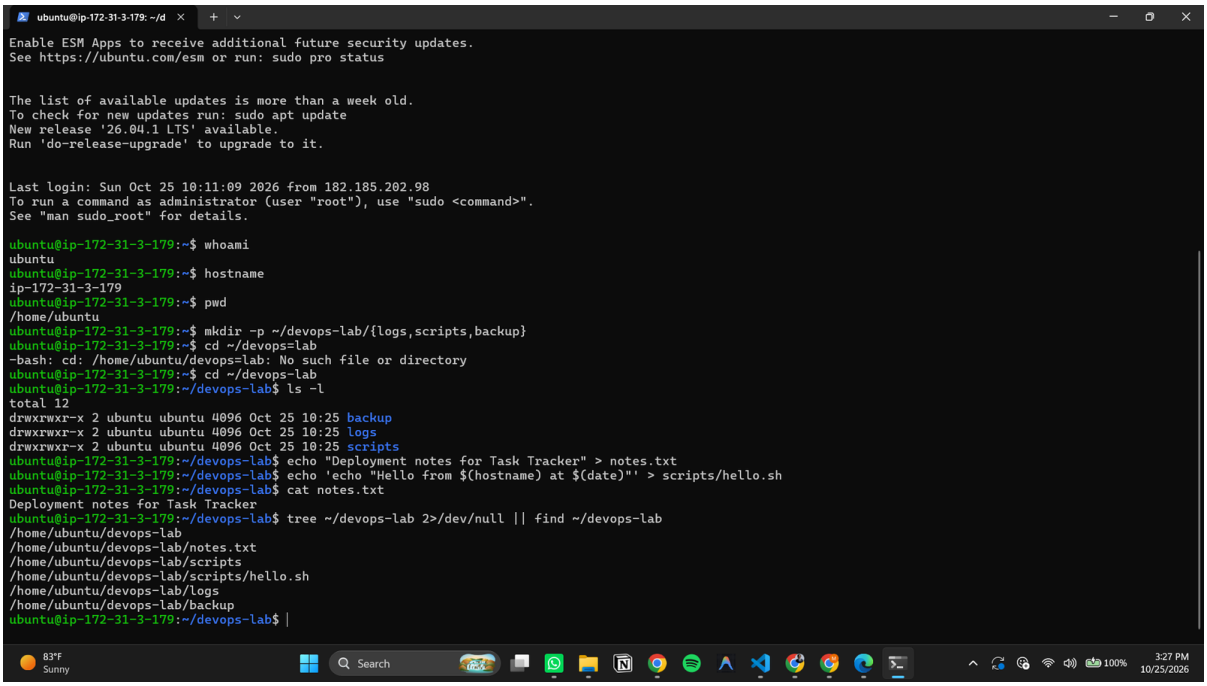

Launch an instance | EC2 | ap
Instances | EC2 | ap-south-1
Auto Scaling group details |
Student Task Tracker
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Instances (6) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
test i-0058994880cb6621a Stopped t3.small – ap-south-1b –
lab-assign-1 i-088261b999e0332f2 Stopped t3.small – ap-south-1b –
task-tracker-asg i-08e26badae8565790 Running t3.micro Initializing ap-south-1a ec2-15-206-75-1.ap-so...
task-tracker-asg i-0eac84e46f351b2ce Terminated t3.micro – ap-south-1a –
Lab-midterm i-026331092ce9a14fb Running t3.medium 3/3 checks passec ap-south-1b ec2-15-252-145-214.ap...
task-tracker-asg i-0b2586384003de901 Running t3.micro 3/3 checks passec ap-south-1b ec2-13-203-198-254.ap...
All instances
Instance alarm states
In alarm
None
Service health
EC2 service status
Operating normally
Health events affecting EC2 instances
Open events
None
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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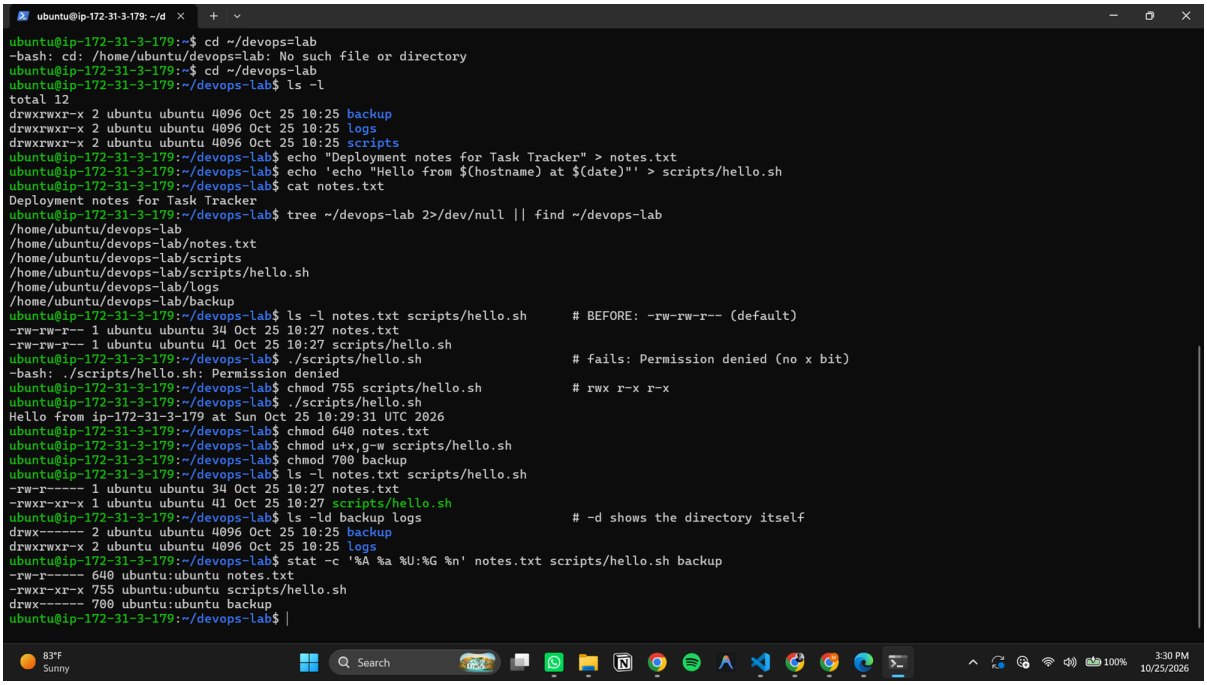

Launch an instance | EC2 | ap
Instances | EC2 | ap-south-1
Auto Scaling group details |
Student Task Tracker
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AutoScalingGroupDetails:id=task-tracker-asg;view=activity
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Auto Scaling groups > task-tracker-asg
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Status
Description
Cause
Start time
Successful
Launching a new EC2 instance: i-08e26badae8565790
At 2026-10-25T11:47:54Z an instance was launched in response to an unhealthy instance needing to be replaced.
2026 October 25, 04:47:55 PM +05:00
Connection draining in progress
Terminating EC2 instance: i-0eac84e46f351b2ce - Waiting For ELB Connection Draining.
At 2026-10-25T11:47:54Z an instance was taken out of service in response to an EC2 health check indicating it has been terminated or stopped.
2026 October 25, 04:47:54 PM +05:00
Successful
Launching a new EC2 instance: i-0eac84e46f351b2ce
At 2026-10-25T11:29:40Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-10-25T11:30:43Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.
2026 October 25, 04:30:45 PM +05:00
Successful
Launching a new EC2 instance: i-0b2586384003de901
At 2026-10-25T11:29:40Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-10-25T11:30:43Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.
2026 October 25, 04:30:45 PM +05:00
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Console Mobile App
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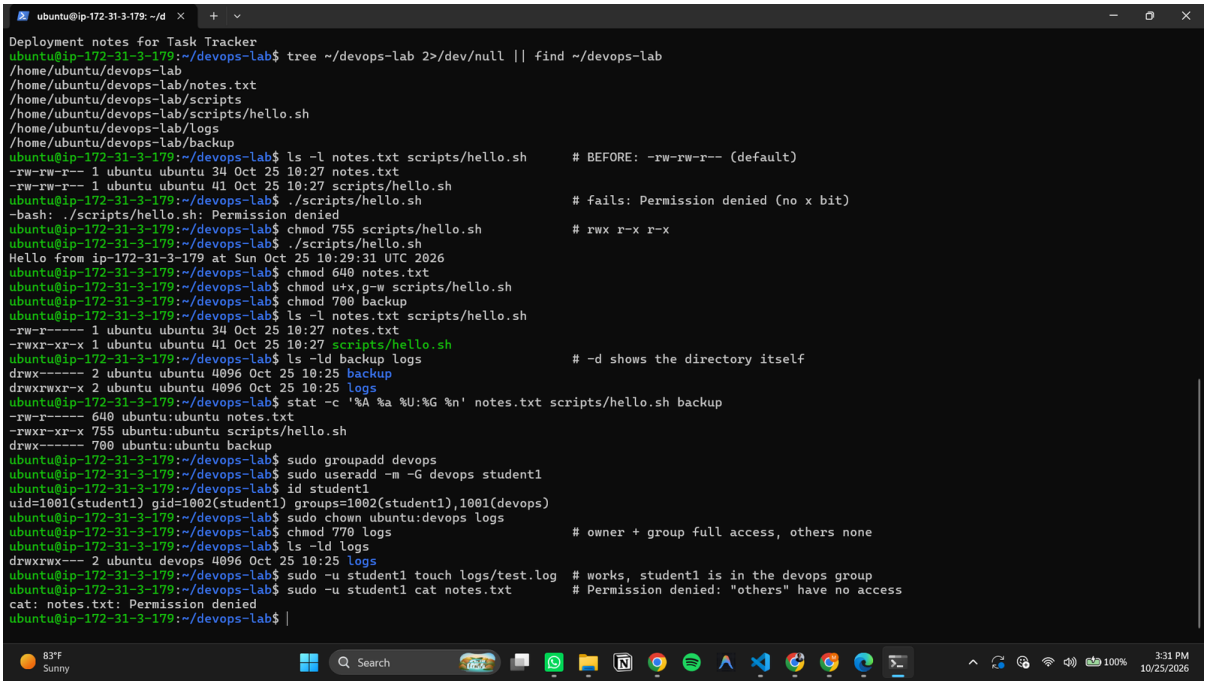

Launch an instance | EC2 | ap
Instances | EC2 | ap-south-1
Edit target group attributes |
Student Task Tracker
Student Task Tracker
Ask Gemini
027551857974-xwllzrqb.ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroupEditAttributes:targetGroupArn=arn:aws:elasticloadbalancing:ap-south-1:027551...
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
aws
Search
[Alt+S]
Asia Pacific (Mumbai)
zayn (027551857974)
root
EC2 > Target groups > task-tracker-tg > Edit target group attributes
0
seconds
30-900 seconds or 0 to disable. Not compatible with the Least outstanding requests and Weighted random routing algorithms.
Target selection configuration
Stickiness | Info
Stickiness allows the load balancer to bind a user's session to a specific target within the target group. The stickiness type differs based on the type of cookie used.
Turn on stickiness
Not compatible with the Weighted random routing algorithm. Can't be turned on if Cross-zone load balancing is off.
Stickiness type
Load balancer generated cookie
Application-based cookie
Stickiness duration
Unit of time
1
days
1 second - 7 days
Cross-zone load balancing | Info
Cross-zone load balancing can be configured for each target group or inherited from the load balancer.
Inherit settings from load balancer attributes
Uses the cross-zone settings from the Application Load Balancer attributes - On by default.
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Task 2: Linux Administration

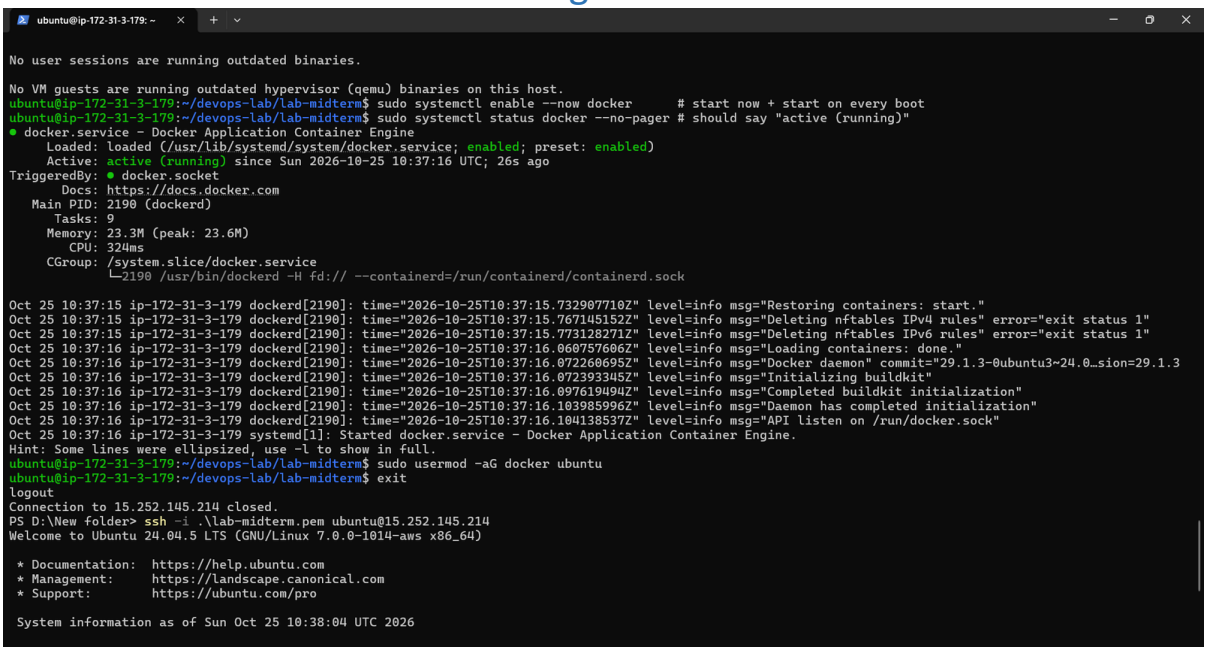





ubuntu@ip-172-31-3-179: ~/d
ubuntu@ip-172-31-3-179:~$ cd ~/devops=lab
-bash: cd: /home/ubuntu/devops=lab: No such file or directory
ubuntu@ip-172-31-3-179:~$ cd ~/devops-lab
ubuntu@ip-172-31-3-179:~/devops-lab$ ls -l
total 12
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 25 10:25 backup
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 25 10:25 logs
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 25 10:25 scripts
ubuntu@ip-172-31-3-179:~/devops-lab$ echo "Deployment notes for Task Tracker" > notes.txt
ubuntu@ip-172-31-3-179:~/devops-lab$ echo 'echo "Hello from $(hostname) at $(date)"' > scripts/hello.sh
ubuntu@ip-172-31-3-179:~/devops-lab$ cat notes.txt
Deployment notes for Task Tracker
ubuntu@ip-172-31-3-179:~/devops-lab$ tree ~/devops-lab 2>/dev/null || find ~/devops-lab
/home/ubuntu/devops-lab
/home/ubuntu/devops-lab/notes.txt
/home/ubuntu/devops-lab/scripts
/home/ubuntu/devops-lab/scripts/hello.sh
/home/ubuntu/devops-lab/logs
/home/ubuntu/devops-lab/backup
ubuntu@ip-172-31-3-179:~/devops-lab$ ls -l notes.txt scripts/hello.sh # BEFORE: -rw-rw-r-- (default)
-rw-rw-r-- 1 ubuntu ubuntu 34 Oct 25 10:27 notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 41 Oct 25 10:27 scripts/hello.sh
ubuntu@ip-172-31-3-179:~/devops-lab$ ./scripts/hello.sh # fails: Permission denied (no x bit)
-bash: ./scripts/hello.sh: Permission denied
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod 755 scripts/hello.sh # rwx r-x r-x
ubuntu@ip-172-31-3-179:~/devops-lab$ ./scripts/hello.sh
Hello from ip-172-31-3-179 at Sun Oct 25 10:29:31 UTC 2026
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod 640 notes.txt
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod u+x,g-w scripts/hello.sh
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod 700 backup
ubuntu@ip-172-31-3-179:~/devops-lab$ ls -l notes.txt scripts/hello.sh
-rw-r----- 1 ubuntu ubuntu 34 Oct 25 10:27 notes.txt
-rwxr-xr-x 1 ubuntu ubuntu 41 Oct 25 10:27 scripts/hello.sh
ubuntu@ip-172-31-3-179:~/devops-lab$ ls -ld backup logs # -d shows the directory itself
drwx------ 2 ubuntu ubuntu 4096 Oct 25 10:25 backup
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 25 10:25 logs
ubuntu@ip-172-31-3-179:~/devops-lab$ stat -c '%A %a %U:%G %n' notes.txt scripts/hello.sh backup
-rw-r----- 640 ubuntu:ubuntu notes.txt
-rwxr-xr-x 755 ubuntu:ubuntu scripts/hello.sh
drwx------ 700 ubuntu:ubuntu backup
ubuntu@ip-172-31-3-179:~/devops-lab$
83°F
Sunny
Search
100%
3:30 PM
10/25/2026


ubuntu@ip-172-31-3-179: ~/d
Deployment notes for Task Tracker
ubuntu@ip-172-31-3-179:~/devops-lab$ tree ~/devops-lab 2>/dev/null || find ~/devops-lab
/home/ubuntu/devops-lab
/home/ubuntu/devops-lab/notes.txt
/home/ubuntu/devops-lab/scripts
/home/ubuntu/devops-lab/scripts/hello.sh
/home/ubuntu/devops-lab/logs
/home/ubuntu/devops-lab/backup
ubuntu@ip-172-31-3-179:~/devops-lab$ ls -l notes.txt scripts/hello.sh # BEFORE: -rw-rw-r-- (default)
-rw-rw-r-- 1 ubuntu ubuntu 34 Oct 25 10:27 notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 41 Oct 25 10:27 scripts/hello.sh
ubuntu@ip-172-31-3-179:~/devops-lab$ ./scripts/hello.sh # fails: Permission denied (no x bit)
-bash: ./scripts/hello.sh: Permission denied
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod 755 scripts/hello.sh # rwx r-x r-x
ubuntu@ip-172-31-3-179:~/devops-lab$ ./scripts/hello.sh
Hello from ip-172-31-3-179 at Sun Oct 25 10:29:31 UTC 2026
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod 640 notes.txt
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod u+x,g-w scripts/hello.sh
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod 700 backup
ubuntu@ip-172-31-3-179:~/devops-lab$ ls -l notes.txt scripts/hello.sh
-rw-r----- 1 ubuntu ubuntu 34 Oct 25 10:27 notes.txt
-rwxr-xr-x 1 ubuntu ubuntu 41 Oct 25 10:27 scripts/hello.sh
ubuntu@ip-172-31-3-179:~/devops-lab$ ls -ld backup logs # -d shows the directory itself
drwx------ 2 ubuntu ubuntu 4096 Oct 25 10:25 backup
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 25 10:25 logs
ubuntu@ip-172-31-3-179:~/devops-lab$ stat -c '%A %a %U:%G %n' notes.txt scripts/hello.sh backup
-rw-r----- 640 ubuntu:ubuntu notes.txt
-rwxr-xr-x 755 ubuntu:ubuntu scripts/hello.sh
drwx------ 700 ubuntu:ubuntu backup
ubuntu@ip-172-31-3-179:~/devops-lab$ sudo groupadd devops
ubuntu@ip-172-31-3-179:~/devops-lab$ sudo useradd -m -G devops student1
ubuntu@ip-172-31-3-179:~/devops-lab$ id student1
uid=1001(student1) gid=1002(student1) groups=1002(student1),1001(devops)
ubuntu@ip-172-31-3-179:~/devops-lab$ sudo chown ubuntu:devops logs
ubuntu@ip-172-31-3-179:~/devops-lab$ chmod 770 logs # owner + group full access, others none
ubuntu@ip-172-31-3-179:~/devops-lab$ ls -ld logs
drwxrwx--- 2 ubuntu devops 4096 Oct 25 10:25 logs
ubuntu@ip-172-31-3-179:~/devops-lab$ sudo -u student1 touch logs/test.log # works, student1 is in the devops group
ubuntu@ip-172-31-3-179:~/devops-lab$ sudo -u student1 cat notes.txt # Permission denied: "others" have no access
cat: notes.txt: Permission denied
ubuntu@ip-172-31-3-179:~/devops-lab$
83°F
Sunny
Search
100%
3:31 PM
10/25/2026

Task 3: Docker Installation & Images



```
ubuntu@ip-172-31-3-179: ~

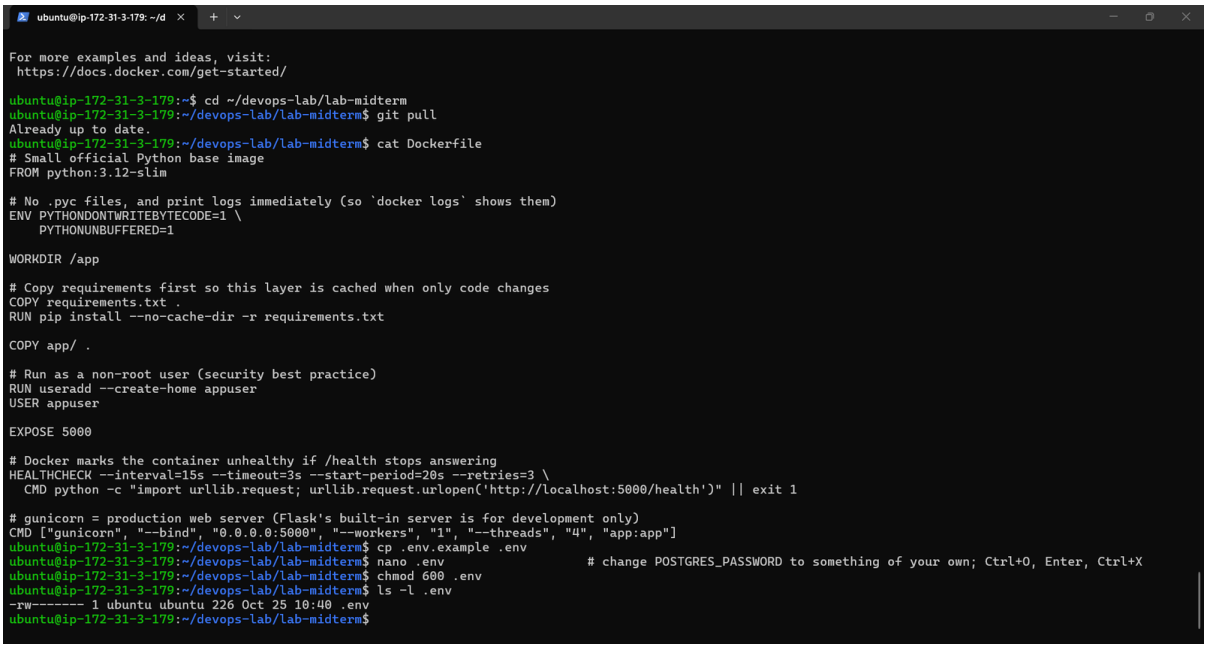
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ sudo systemctl enable --now docker      # start now + start on every boot
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ sudo systemctl status docker --no-pager # should say "active (running)"
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-10-25 10:37:16 UTC; 26s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2190 (dockerd)
      Tasks: 9
     Memory: 23.3M (peak: 23.6M)
        CPU: 324ms
     CGroup: /system.slice/docker.service
             └─2190 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 25 10:37:15 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:15.732907710Z" level=info msg="Restoring containers: start."
Oct 25 10:37:15 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:15.767145152Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Oct 25 10:37:15 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:15.773128271Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Oct 25 10:37:16 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:16.060757606Z" level=info msg="Loading containers: done."
Oct 25 10:37:16 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:16.072260695Z" level=info msg="Docker daemon" commit="29.1.3-0ubuntu3~24.0…sion=29.1.3
Oct 25 10:37:16 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:16.072393345Z" level=info msg="Initializing buildkit"
Oct 25 10:37:16 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:16.097619494Z" level=info msg="Completed buildkit initialization"
Oct 25 10:37:16 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:16.103985996Z" level=info msg="Daemon has completed initialization"
Oct 25 10:37:16 ip-172-31-3-179 dockerd[2190]: time="2026-10-25T10:37:16.104138537Z" level=info msg="API listen on /run/docker.sock"
Oct 25 10:37:16 ip-172-31-3-179 systemd[1]: Started docker.service - Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ exit
logout
Connection to 15.252.145.214 closed.
PS D:\New folder> ssh -i .\lab-midterm.pem ubuntu@15.252.145.214
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Oct 25 10:38:04 UTC 2026
```



```
ubuntu@ip-172-31-3-179: ~/d

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

ubuntu@ip-172-31-3-179:~$ cd ~/devops-lab/lab-midterm
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ git pull
Already up to date.
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ cat Dockerfile
# Small official Python base image
FROM python:3.12-slim

# No .pyc files, and print logs immediately (so `docker logs` shows them)
ENV PYTHONDONTWRITEBYTECODE=1 \
    PYTHONUNBUFFERED=1

WORKDIR /app

# Copy requirements first so this layer is cached when only code changes
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

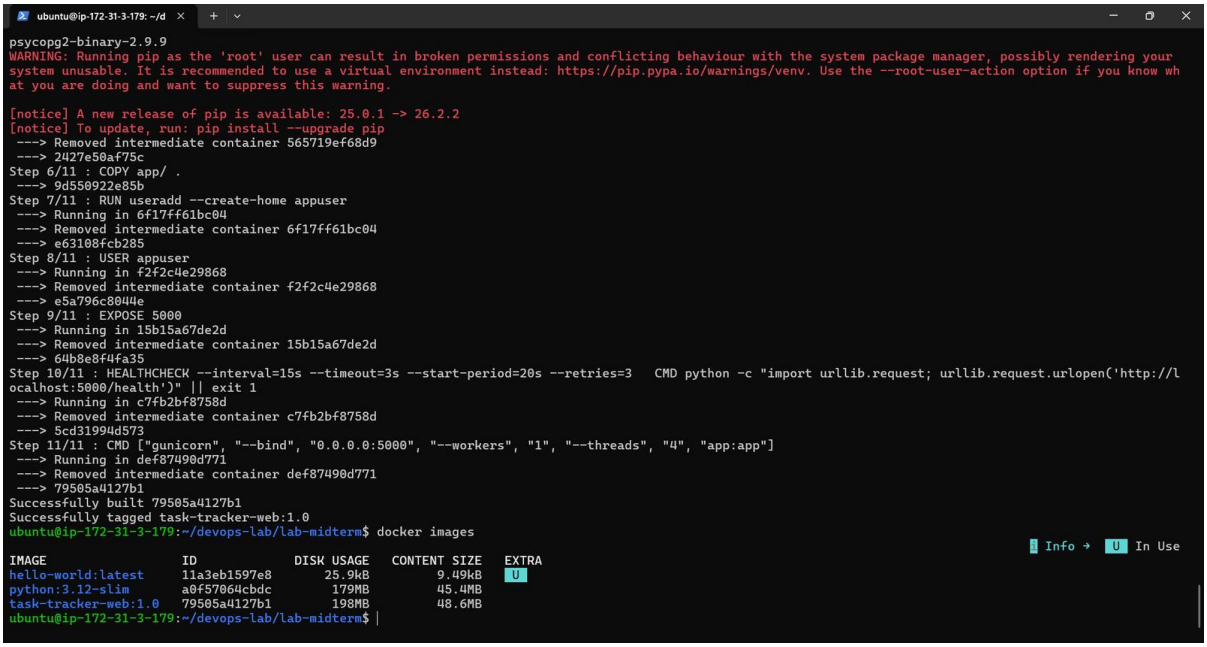
COPY app/ .

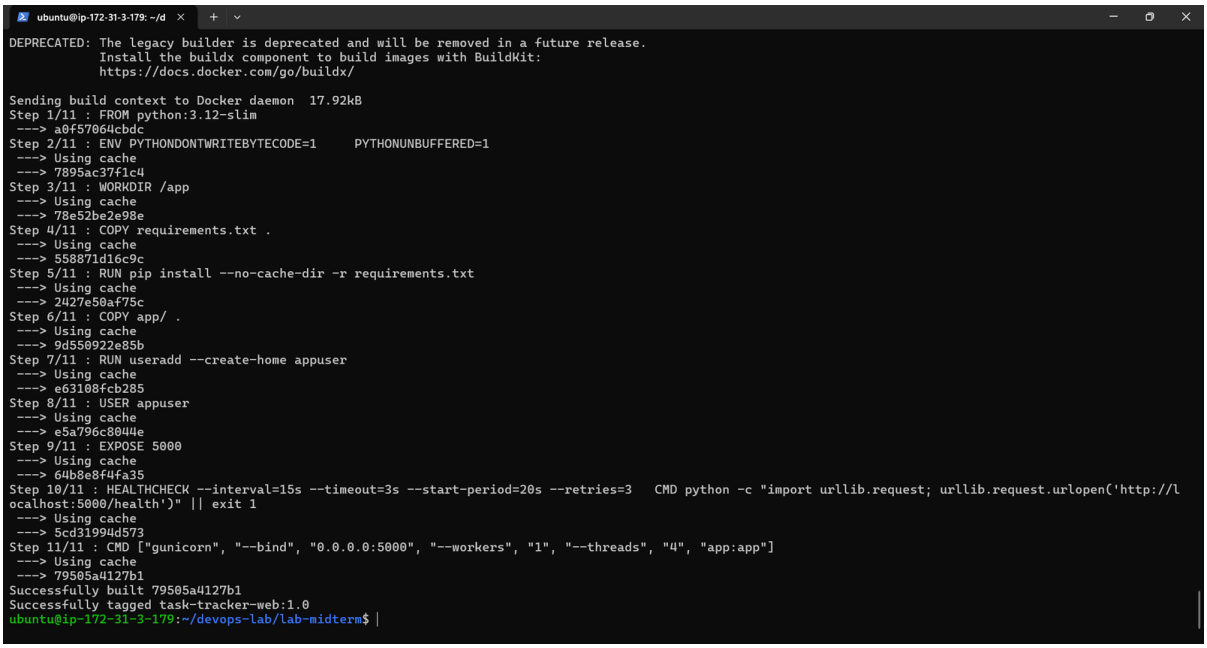
# Run as a non-root user (security best practice)
RUN useradd --create-home appuser
USER appuser

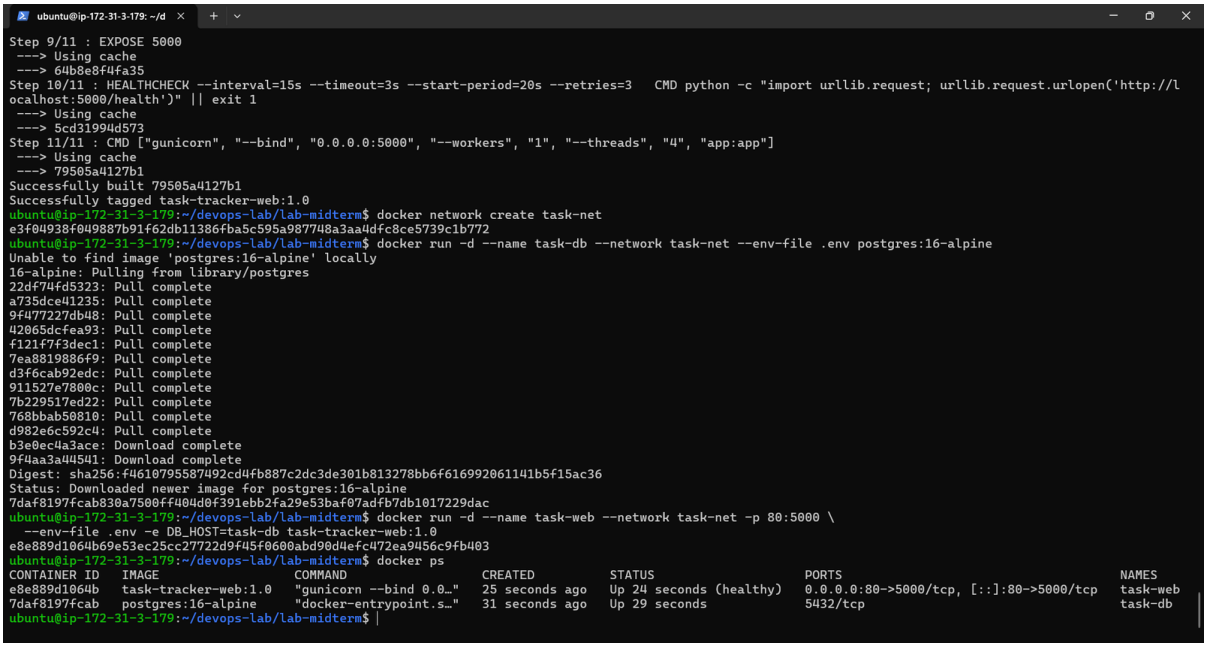
EXPOSE 5000

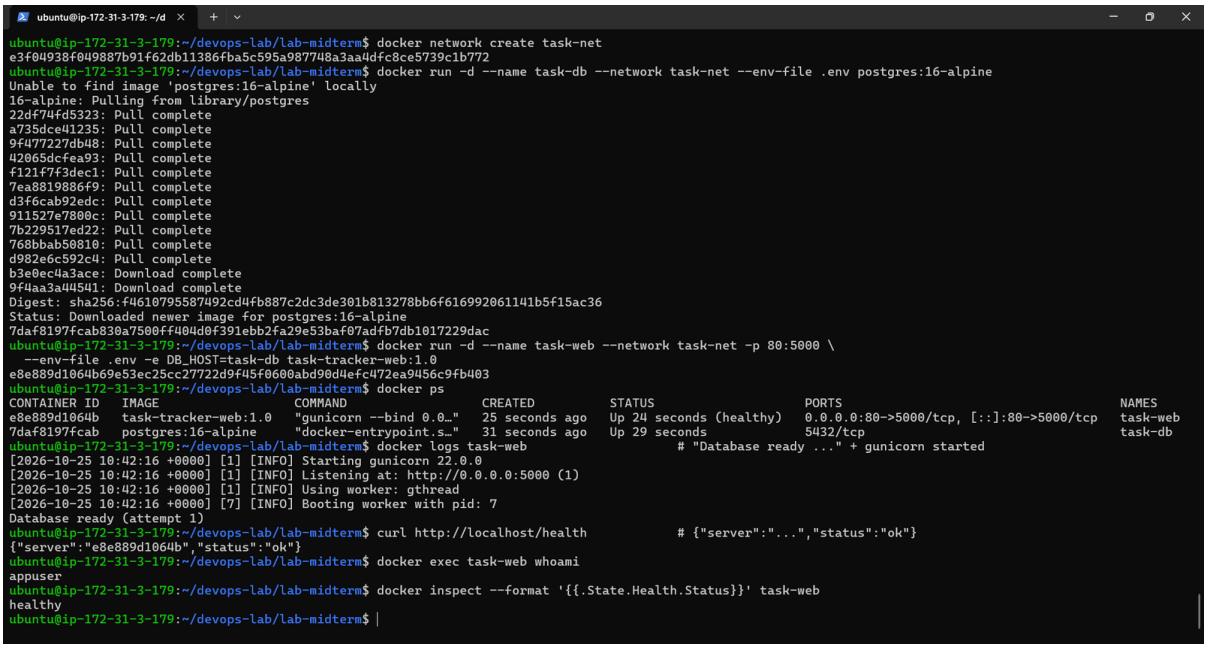
# Docker marks the container unhealthy if /health stops answering
HEALTHCHECK --interval=15s --timeout=3s --start-period=20s --retries=3 \
  CMD python -c "import urllib.request; urllib.request.urlopen('http://localhost:5000/health')" || exit 1

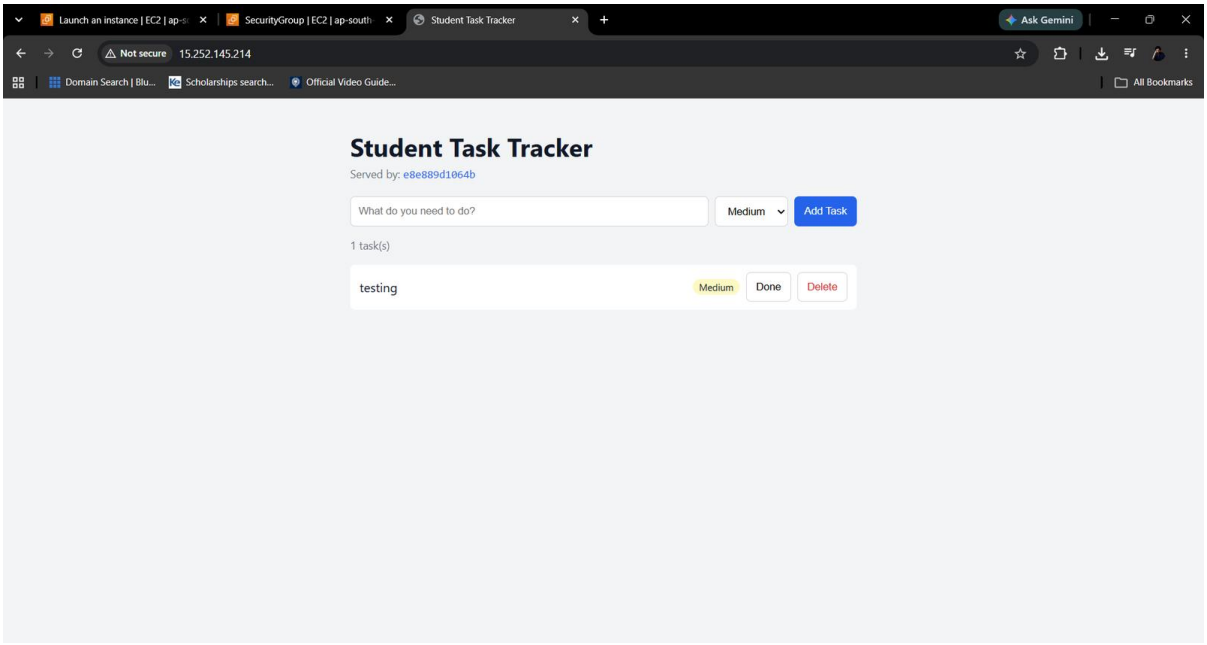
# gunicorn = production web server (Flask's built-in server is for development only)
CMD ["gunicorn", "--bind", "0.0.0.0:5000", "--workers", "1", "--threads", "4", "app:app"]
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ cp .env.example .env
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ nano .env                  # change POSTGRES_PASSWORD to something of your own; Ctrl+O, Enter, Ctrl+X
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ chmod 600 .env
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ ls -l .env
-rw------- 1 ubuntu ubuntu 226 Oct 25 10:40 .env
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$
```


ubuntu@ip-172-31-3-179: ~/d
psycopg2-binary-2.9.9
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv. Use the --root-user-action option if you know wh at you are doing and want to suppress this warning.
[notice] A new release of pip is available: 25.0.1 -> 26.2.2
[notice] To update, run: pip install --upgrade pip
---> Removed intermediate container 565719ef68d9
---> 2427e50af75c
Step 6/11 : COPY app/ .
---> 9d550922e85b
Step 7/11 : RUN useradd --create-home appuser
---> Running in 6f17ff61bc04
---> Removed intermediate container 6f17ff61bc04
---> e63108fcb285
Step 8/11 : USER appuser
---> Running in f2f2c4e29868
---> Removed intermediate container f2f2c4e29868
---> e5a796c8044e
Step 9/11 : EXPOSE 5000
---> Running in 15b15a67de2d
---> Removed intermediate container 15b15a67de2d
---> 64b8e8f4fa35
Step 10/11 : HEALTHCHECK --interval=15s --timeout=3s --start-period=20s --retries=3 CMD python -c "import urllib.request; urllib.request.urlopen('http://l ocalhost:5000/health')" || exit 1
---> Running in c7fb2bf8758d
---> Removed intermediate container c7fb2bf8758d
---> 5cd31994d573
Step 11/11 : CMD ["gunicorn", "--bind", "0.0.0.0:5000", "--workers", "1", "--threads", "4", "app:app"]
---> Running in def87490d771
---> Removed intermediate container def87490d771
---> 79505a4127b1
Successfully built 79505a4127b1
Successfully tagged task-tracker-web:1.0
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker images
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
hello-world:latest 11a3eb1597e8 25.9kB 9.49kB U
python:3.12-slim a0f57064cbdc 179MB 45.4MB
task-tracker-web:1.0 79505a4127b1 198MB 48.6MB
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$

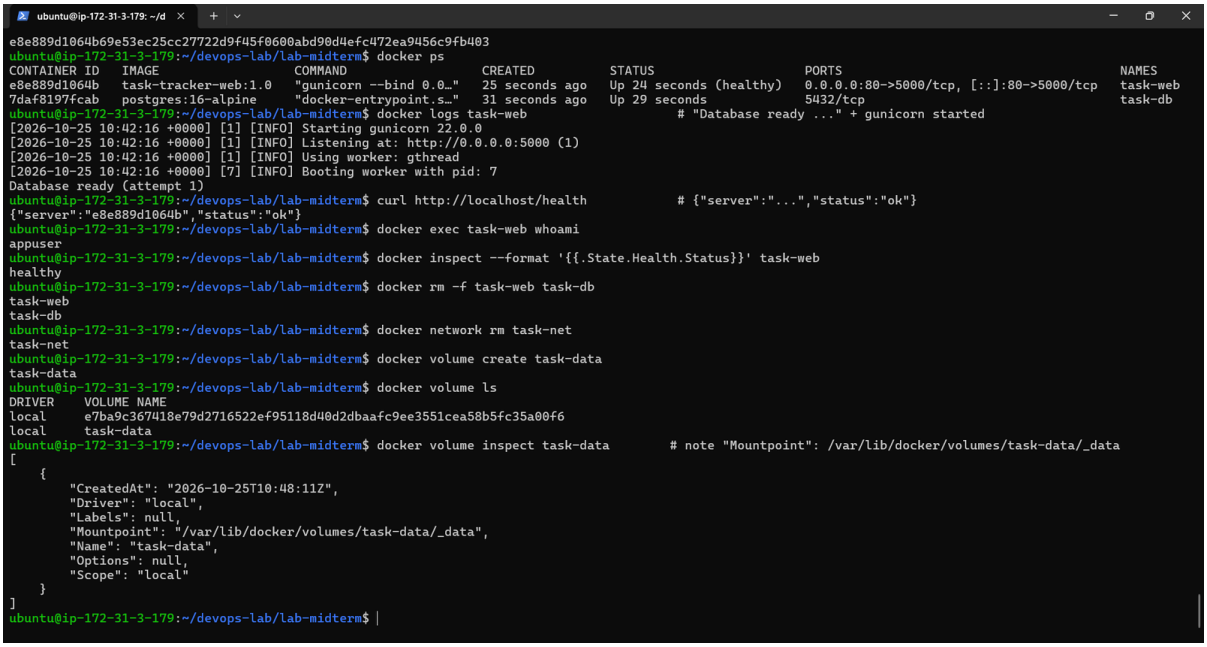

ubuntu@ip-172-31-3-179: ~/d
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/
Sending build context to Docker daemon 17.92kB
Step 1/11 : FROM python:3.12-slim
---> a0f57064cbdc
Step 2/11 : ENV PYTHONDONTWRITEBYTECODE=1 PYTHONUNBUFFERED=1
---> Using cache
---> 7895ac37f1c4
Step 3/11 : WORKDIR /app
---> Using cache
---> 78e52be2e98e
Step 4/11 : COPY requirements.txt .
---> Using cache
---> 558871d16c9c
Step 5/11 : RUN pip install --no-cache-dir -r requirements.txt
---> Using cache
---> 2427e50af75c
Step 6/11 : COPY app/ .
---> Using cache
---> 9d550922e85b
Step 7/11 : RUN useradd --create-home appuser
---> Using cache
---> e63108fcb285
Step 8/11 : USER appuser
---> Using cache
---> e5a796c8044e
Step 9/11 : EXPOSE 5000
---> Using cache
---> 64b8e8f4fa35
Step 10/11 : HEALTHCHECK --interval=15s --timeout=3s --start-period=20s --retries=3 CMD python -c "import urllib.request; urllib.request.urlopen('http://l ocalhost:5000/health')" || exit 1
---> Using cache
---> 5cd31994d573
Step 11/11 : CMD ["gunicorn", "--bind", "0.0.0.0:5000", "--workers", "1", "--threads", "4", "app:app"]
---> Using cache
---> 79505a4127b1
Successfully built 79505a4127b1
Successfully tagged task-tracker-web:1.0
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$


ubuntu@ip-172-31-3-179: ~/d
Step 9/11 : EXPOSE 5000
 ---> Using cache
 ---> 64b8e8f4fa35
Step 10/11 : HEALTHCHECK --interval=15s --timeout=3s --start-period=20s --retries=3   CMD python -c "import urllib.request; urllib.request.urlopen('http://localhost:5000/health')" || exit 1
 ---> Using cache
 ---> 5cd31994d573
Step 11/11 : CMD ["gunicorn", "--bind", "0.0.0.0:5000", "--workers", "1", "--threads", "4", "app:app"]
 ---> Using cache
 ---> 79505a4127b1
Successfully built 79505a4127b1
Successfully tagged task-tracker-web:1.0
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network create task-net
e3f04938f049887b91f62db11386fba5c595a987748a3aa4dfc8ce5739c1b772
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-db --network task-net --env-file .env postgres:16-alpine
Unable to find image 'postgres:16-alpine' locally
16-alpine: Pulling from library/postgres
22df74fd5323: Pull complete
a735dce41235: Pull complete
9f477227db48: Pull complete
42065dcfea93: Pull complete
f121f7f3dec1: Pull complete
7ea8819886f9: Pull complete
d3f6cab92edc: Pull complete
911527e7800c: Pull complete
7b229517ed22: Pull complete
768bbab50810: Pull complete
d982e6c592c4: Pull complete
b3e0ec4a3ace: Download complete
9f4aa3a44541: Download complete
Digest: sha256:f46107955874 92cd4fb887c2dc3de301b813278bb6f616992061141b5f15ac36
Status: Downloaded newer image for postgres:16-alpine
7daf8197fcab830a7500ff404d0f391ebb2fa29e53baf07adfb7db1017229dac
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-web --network task-net -p 80:5000 \
  --env-file .env -e DB_HOST=task-db task-tracker-web:1.0
e8e889d1064b69e53ec25cc27722d9f45f0600abd90d4efc472ea9456c9fb403
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps
CONTAINER ID   IMAGE   COMMAND   CREATED   STATUS   PORTS   NAMES
e8e889d1064b   task-tracker-web:1.0   "gunicorn --bind 0.0…"   25 seconds ago   Up 24 seconds (healthy)   0.0.0.0:80->5000/tcp, [::]:80->5000/tcp   task-web
7daf8197fcab   postgres:16-alpine   "docker-entrypoint.s…"   31 seconds ago   Up 29 seconds   5432/tcp   task-db
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$

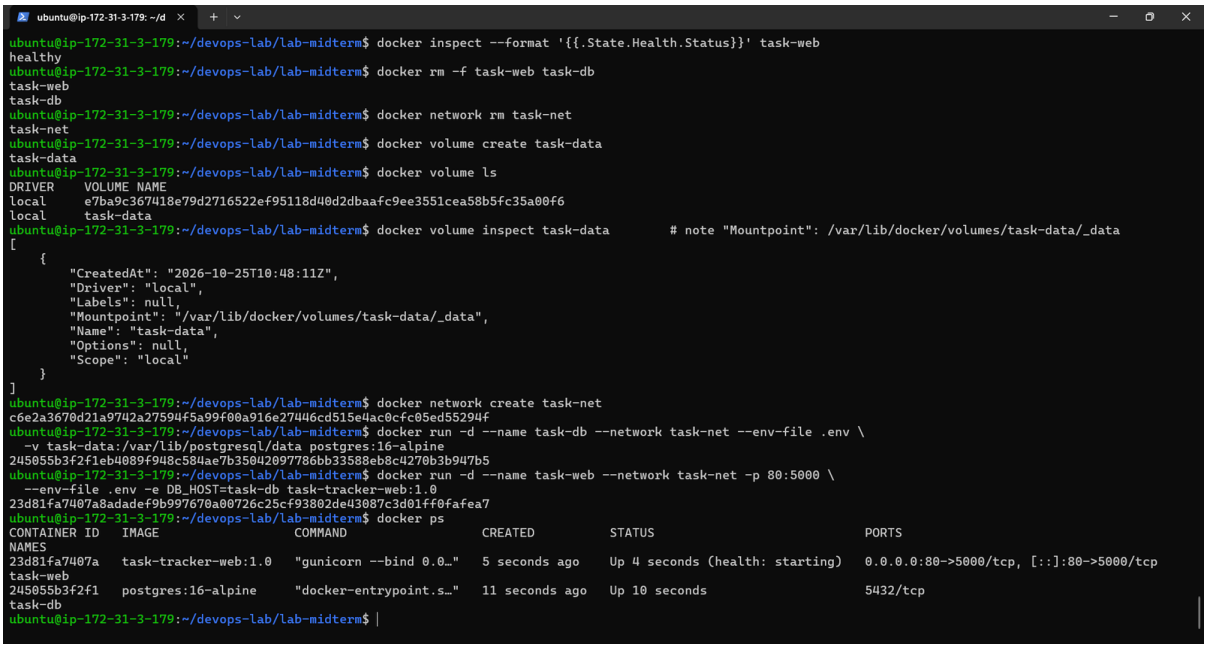

ubuntu@ip-172-31-3-179: ~/d
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network create task-net
e3f04938f049887b91f62db11386fba5c595a987748a3aa4dfc8ce5739c1b772
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-db --network task-net --env-file .env postgres:16-alpine
Unable to find image 'postgres:16-alpine' locally
16-alpine: Pulling from library/postgres
22df74fd5323: Pull complete
a735dce41235: Pull complete
9f477227db48: Pull complete
42065dcfea93: Pull complete
f121f7f3dec1: Pull complete
7ea8819886f9: Pull complete
d3f6cab92edc: Pull complete
911527e7800c: Pull complete
7b229517ed22: Pull complete
768bbab50810: Pull complete
d982e6c592c4: Pull complete
b3e0ec4a3ace: Download complete
9f4aa3a44541: Download complete
Digest: sha256:f46107955874 92cd4fb887c2dc3de301b813278bb6f616992061141b5f15ac36
Status: Downloaded newer image for postgres:16-alpine
7daf8197fcab830a7500ff404d0f391ebb2fa29e53baf07adfb7db1017229dac
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-web --network task-net -p 80:5000 \
  --env-file .env -e DB_HOST=task-db task-tracker-web:1.0
e8e889d1064b69e53ec25cc27722d9f45f0600abd90d4efc472ea9456c9fb403
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps
CONTAINER ID   IMAGE   COMMAND   CREATED   STATUS   PORTS   NAMES
e8e889d1064b   task-tracker-web:1.0   "gunicorn --bind 0.0…"   25 seconds ago   Up 24 seconds (healthy)   0.0.0.0:80->5000/tcp, [::]:80->5000/tcp   task-web
7daf8197fcab   postgres:16-alpine   "docker-entrypoint.s…"   31 seconds ago   Up 29 seconds   5432/tcp   task-db
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker logs task-web   # "Database ready ..." + gunicorn started
[2026-10-25 10:42:16 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-10-25 10:42:16 +0000] [1] [INFO] Listening at: http://0.0.0.0:5000 (1)
[2026-10-25 10:42:16 +0000] [1] [INFO] Using worker: gthread
[2026-10-25 10:42:16 +0000] [7] [INFO] Booting worker with pid: 7
Database ready (attempt 1)
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ curl http://localhost/health   # {"server":"...","status":"ok"}
{"server":"e8e889d1064b","status":"ok"}
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker exec task-web whoami
appuser
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker inspect --format '{{.State.Health.Status}}' task-web
healthy
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$


Launch an instance | EC2 | ap-s
SecurityGroup | EC2 | ap-south
Student Task Tracker
Ask Gemini
Not secure
15.252.145.214
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Student Task Tracker
Served by: e8e889d1064b
What do you need to do?
Medium
Add Task
1 task(s)
testing
Medium
Done
Delete

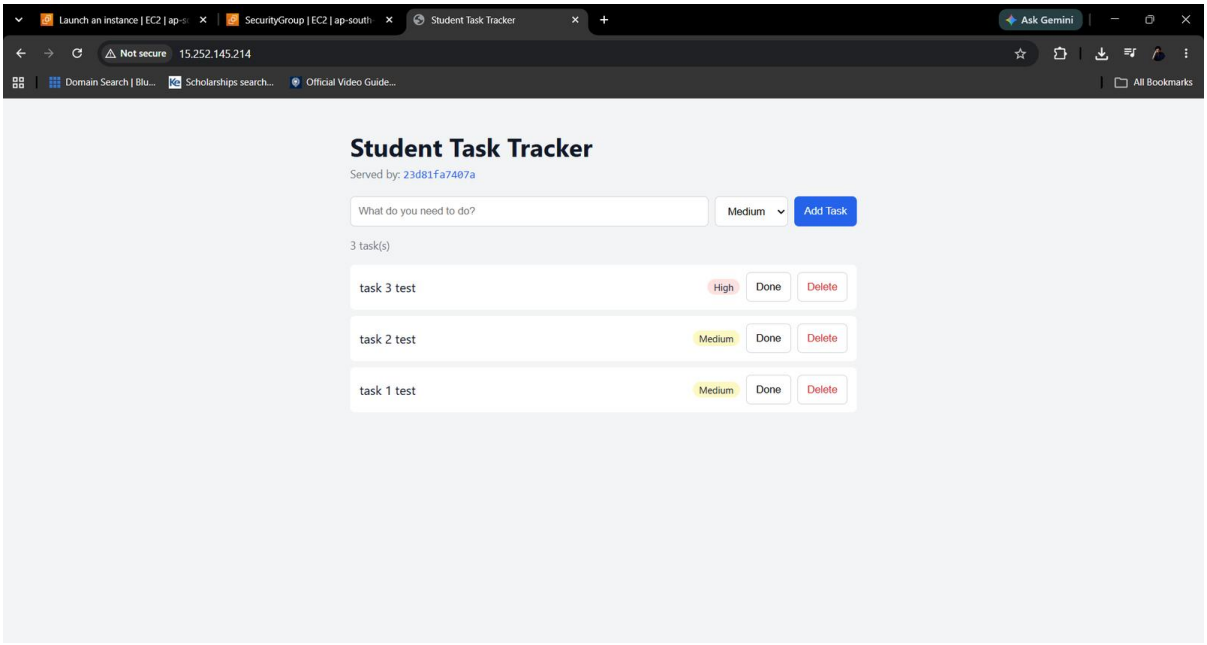
Task 4: Docker Volumes

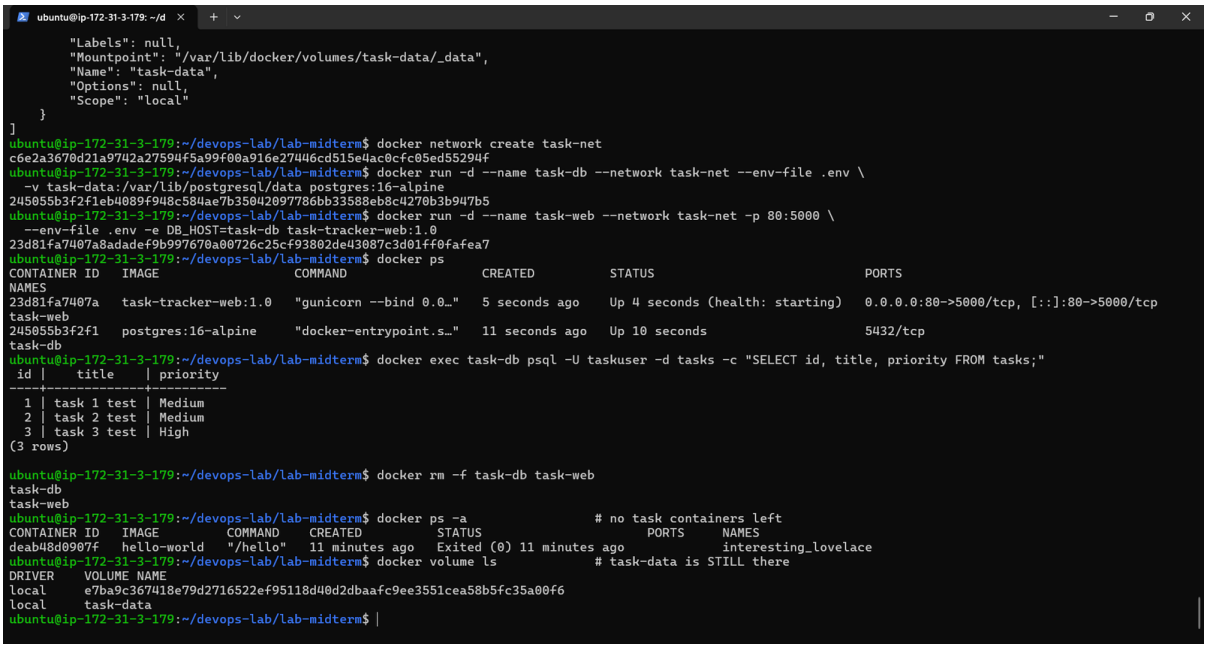


```
e8e889d1064b69e53ec25cc27722d9f45f0600abd90d4efc472ea9456c9fb403
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS                    PORTS                                       NAMES
e8e889d1064b   task-tracker-web:1.0   "gunicorn --bind 0.0…"   25 seconds ago   Up 24 seconds (healthy)   0.0.0.0:80->5000/tcp, [::]:80->5000/tcp   task-web
7daf8197fcab   postgres:16-alpine     "docker-entrypoint.s…"   31 seconds ago   Up 29 seconds             5432/tcp                                    task-db
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker logs task-web                  # "Database ready ..." + gunicorn started
[2026-10-25 10:42:16 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-10-25 10:42:16 +0000] [1] [INFO] Listening at: http://0.0.0.0:5000 (1)
[2026-10-25 10:42:16 +0000] [1] [INFO] Using worker: gthread
[2026-10-25 10:42:16 +0000] [7] [INFO] Booting worker with pid: 7
Database ready (attempt 1)
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ curl http://localhost/health           # {"server":"...","status":"ok"}
{"server":"e8e889d1064b","status":"ok"}
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker exec task-web whoami
appuser
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker inspect --format '{{.State.Health.Status}}' task-web
healthy
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker rm -f task-web task-db
task-web
task-db
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network rm task-net
task-net
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume create task-data
task-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume ls
DRIVER    VOLUME NAME
local     e7ba9c367418e79d2716522ef95118d40d2dbaafc9ee3551cea58b5fc35a00f6
local     task-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume inspect task-data        # note "Mountpoint": /var/lib/docker/volumes/task-data/_data
[
    {
        "CreatedAt": "2026-10-25T10:48:11Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/task-data/_data",
        "Name": "task-data",
        "Options": null,
        "Scope": "local"
    }
]
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$
```



```
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker inspect --format '{{.State.Health.Status}}' task-web
healthy
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker rm -f task-web task-db
task-web
task-db
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network rm task-net
task-net
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume create task-data
task-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume ls
DRIVER    VOLUME NAME
local     e7ba9c367418e79d2716522ef95118d40d2dbaafc9ee3551cea58b5fc35a00f6
local     task-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume inspect task-data        # note "Mountpoint": /var/lib/docker/volumes/task-data/_data
[
    {
        "CreatedAt": "2026-10-25T10:48:11Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/task-data/_data",
        "Name": "task-data",
        "Options": null,
        "Scope": "local"
    }
]
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network create task-net
c6e2a3670d21a9742a27594f5a99f00a916e27446cd515e4ac0cfc05ed55294f
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-db --network task-net --env-file .env \
  -v task-data:/var/lib/postgresql/data postgres:16-alpine
245055b3f2f1eb4089f948c584ae7b35042097786bb33588eb8c4270b3b947b5
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-web --network task-net -p 80:5000 \
  --env-file .env -e DB_HOST=task-db task-tracker-web:1.0
23d81fa7407a8adadef9b997670a00726c25cf93802de43087c3d01ff0fafea7
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS                            PORTS                                       NAMES
23d81fa7407a   task-tracker-web:1.0   "gunicorn --bind 0.0…"   5 seconds ago    Up 4 seconds (health: starting)   0.0.0.0:80->5000/tcp, [::]:80->5000/tcp   task-web
245055b3f2f1   postgres:16-alpine     "docker-entrypoint.s…"   11 seconds ago   Up 10 seconds                     5432/tcp                                    task-db
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$
```


Launch an instance | EC2 | ap-s
SecurityGroup | EC2 | ap-south
Student Task Tracker
Ask Gemini
Not secure
15.252.145.214
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Student Task Tracker
Served by: 23d81fa7407a
What do you need to do?
Medium
Add Task
3 task(s)
task 3 test
High
Done
Delete
task 2 test
Medium
Done
Delete
task 1 test
Medium
Done
Delete


ubuntu@ip-172-31-3-179: ~/d
"Labels": null,
"Mountpoint": "/var/lib/docker/volumes/task-data/_data",
"Name": "task-data",
"Options": null,
"Scope": "local"
}
]
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network create task-net
c6e2a3670d21a9742a27594f5a99f00a916e27446cd515e4ac0cfc05ed55294f
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-db --network task-net --env-file .env \
-v task-data:/var/lib/postgresql/data postgres:16-alpine
245055b3f2f1eb4089f948c584ae7b35042097786bb33588eb8c4270b3b947b5
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-web --network task-net -p 80:5000 \
--env-file .env -e DB_HOST=task-db task-tracker-web:1.0
23d81fa7407a8adadef9b997670a00726c25cf93802de43087c3d01ff0fafea7
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
23d81fa7407a task-tracker-web:1.0 "gunicorn --bind 0.0…" 5 seconds ago Up 4 seconds (health: starting) 0.0.0.0:80->5000/tcp, [::]:80->5000/tcp
task-web
245055b3f2f1 postgres:16-alpine "docker-entrypoint.s…" 11 seconds ago Up 10 seconds 5432/tcp
task-db
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker exec task-db psql -U taskuser -d tasks -c "SELECT id, title, priority FROM tasks;"
id | title | priority
1 | task 1 test | Medium
2 | task 2 test | Medium
3 | task 3 test | High
(3 rows)
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker rm -f task-db task-web
task-db
task-web
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps -a # no task containers left
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
deab48d0907f hello-world "/hello" 11 minutes ago Exited (0) 11 minutes ago interesting_lovelace
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume ls # task-data is STILL there
DRIVER VOLUME NAME
local e7ba9c367418e79d2716522ef95118d40d2dbaafc9ee3551cea58b5fc35a00f6
local task-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$

```
ubuntu@ip-172-31-3-179: ~/d
23d81fa7407a8adadef9b997670a00726c25cf93802de43087c3d01ff0fafea7
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS                            PORTS
NAMES
23d81fa7407a   task-tracker-web:1.0   "gunicorn --bind 0.0…"   5 seconds ago    Up 4 seconds (health: starting)   0.0.0.0:80->5000/tcp, [::]:80->5000/tcp
task-web
245055b3f2f1   postgres:16-alpine     "docker-entrypoint.s…"   11 seconds ago   Up 10 seconds                     5432/tcp
task-db
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker exec task-db psql -U taskuser -d tasks -c "SELECT id, title, priority FROM tasks;"
 id |    title    | priority
----+-------------+----------
  1 | task 1 test | Medium
  2 | task 2 test | Medium
  3 | task 3 test | High
(3 rows)

ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker rm -f task-db task-web
task-db
task-web
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps -a                # no task containers left
CONTAINER ID   IMAGE         COMMAND    CREATED          STATUS                      PORTS     NAMES
deab48d0907f   hello-world   "/hello"   11 minutes ago   Exited (0) 11 minutes ago             interesting_lovelace
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume ls            # task-data is STILL there
DRIVER    VOLUME NAME
local     e7ba9c367418e79d2716522ef95118d40d2dbaafc9ee3551cea58b5fc35a00f6
local     task-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-db --network task-net --env-file .env \
  -v task-data:/var/lib/postgresql/data postgres:16-alpine
743cba0106bd540bee5d7d992f4943c3aa53d4ed115d3214f39cf349d6552ff5
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-web --network task-net -p 80:5000 \
  --env-file .env -e DB_HOST=task-db task-tracker-web:1.0
c7d036cb7a3f0df03be81c0a9140d19fc6b658a91200d6f77e7774960303241c
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker exec task-db psql -U taskuser -d tasks -c "SELECT id, title, priority FROM tasks;"
 id |    title    | priority
----+-------------+----------
  1 | task 1 test | Medium
  2 | task 2 test | Medium
  3 | task 3 test | High
(3 rows)

ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$
```

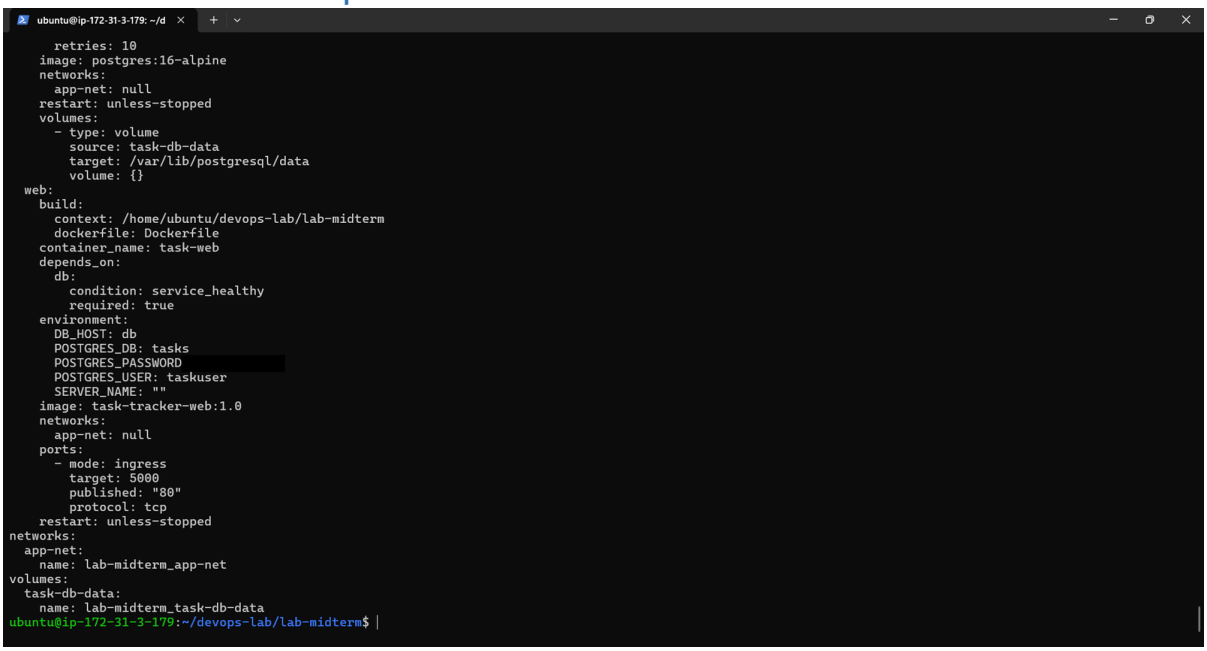
```
ubuntu@ip-172-31-3-179: ~/d
  3 | task 3 test | High
(3 rows)

ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker rm -f task-db task-web
task-db
task-web
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker ps -a                # no task containers left
CONTAINER ID   IMAGE         COMMAND    CREATED          STATUS                      PORTS     NAMES
deab48d0907f   hello-world   "/hello"   11 minutes ago   Exited (0) 11 minutes ago             interesting_lovelace
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume ls            # task-data is STILL there
DRIVER    VOLUME NAME
local     e7ba9c367418e79d2716522ef95118d40d2dbaafc9ee3551cea58b5fc35a00f6
local     task-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-db --network task-net --env-file .env \
  -v task-data:/var/lib/postgresql/data postgres:16-alpine
743cba0106bd540bee5d7d992f4943c3aa53d4ed115d3214f39cf349d6552ff5
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker run -d --name task-web --network task-net -p 80:5000 \
  --env-file .env -e DB_HOST=task-db task-tracker-web:1.0
c7d036cb7a3f0df03be81c0a9140d19fc6b658a91200d6f77e7774960303241c
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker exec task-db psql -U taskuser -d tasks -c "SELECT id, title, priority FROM tasks;"
 id |    title    | priority
----+-------------+----------
  1 | task 1 test | Medium
  2 | task 2 test | Medium
  3 | task 3 test | High
(3 rows)

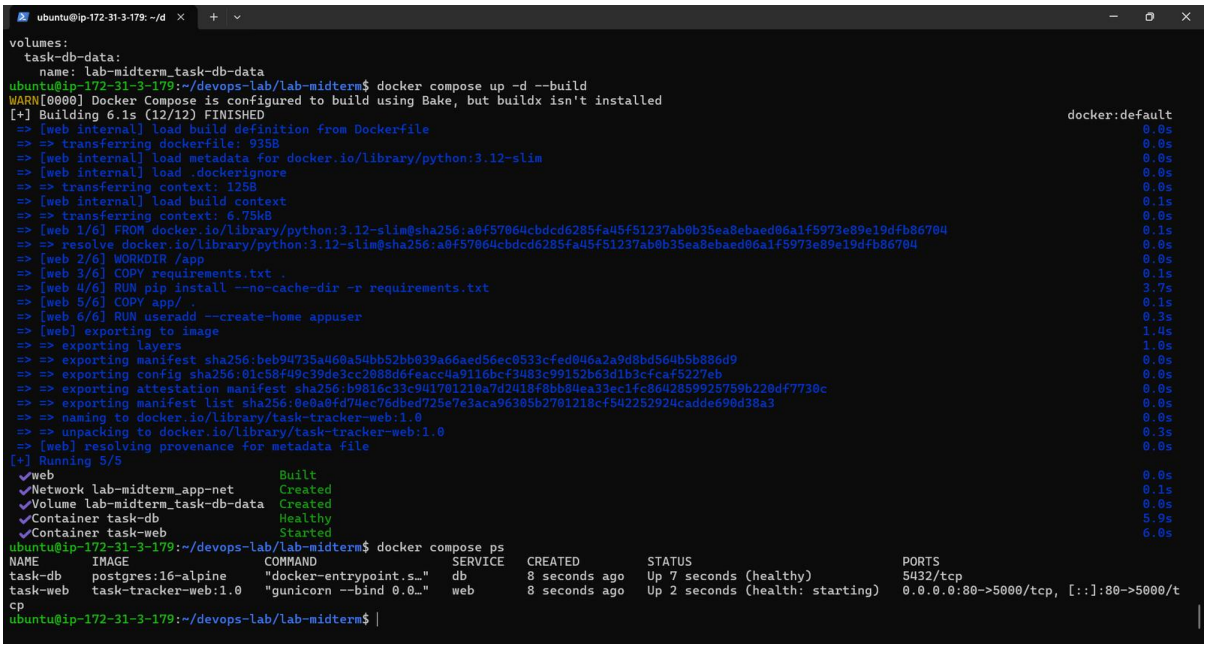
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ sudo ls /var/lib/docker/volumes/task-data/_data
PG_VERSION  pg_commit_ts  pg_ident.conf  pg_notify    pg_snapshots  pg_subtrans  pg_wal               postgresql.conf
base        pg_dynshmem   pg_logical     pg_replslot  pg_stat       pg_tblspc    pg_xact              postmaster.opts
global      pg_hba.conf   pg_multixact   pg_serial    pg_stat_tmp   pg_twophase  postgresql.auto.conf  postmaster.pid
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker rm -f task-db task-web
task-db
task-web
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network rm task-net
task-net
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume rm task-data   # a volume is only deleted when you explicitly remove it


task-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$
```

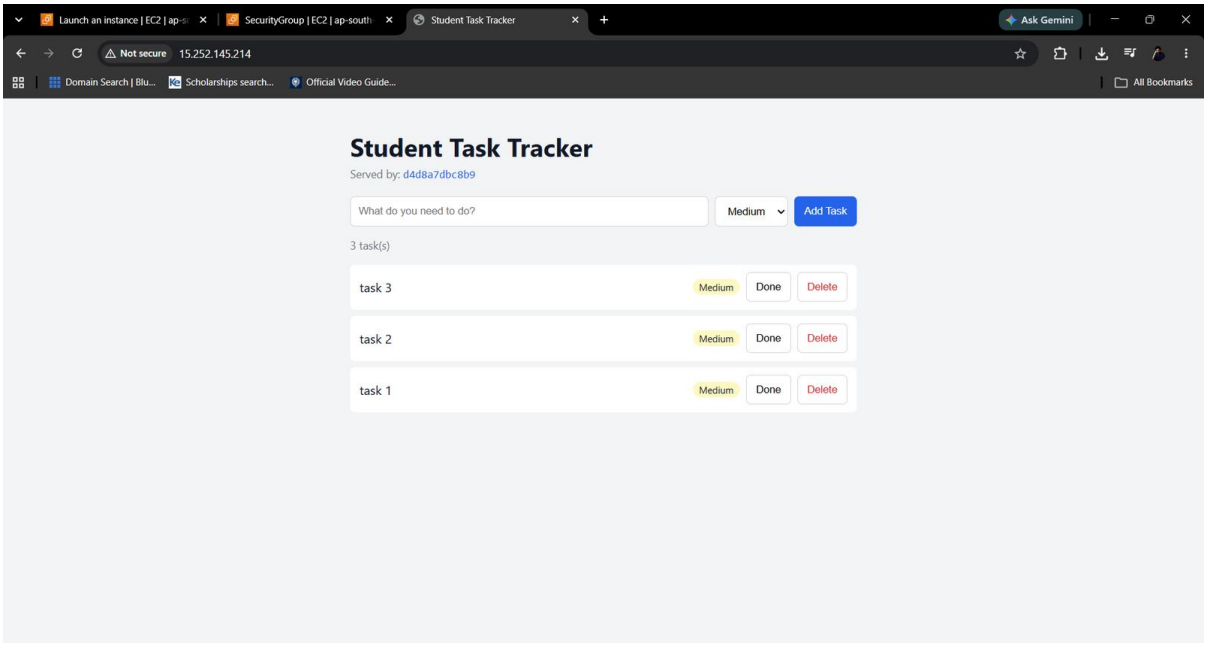
Task 5: Docker Compose

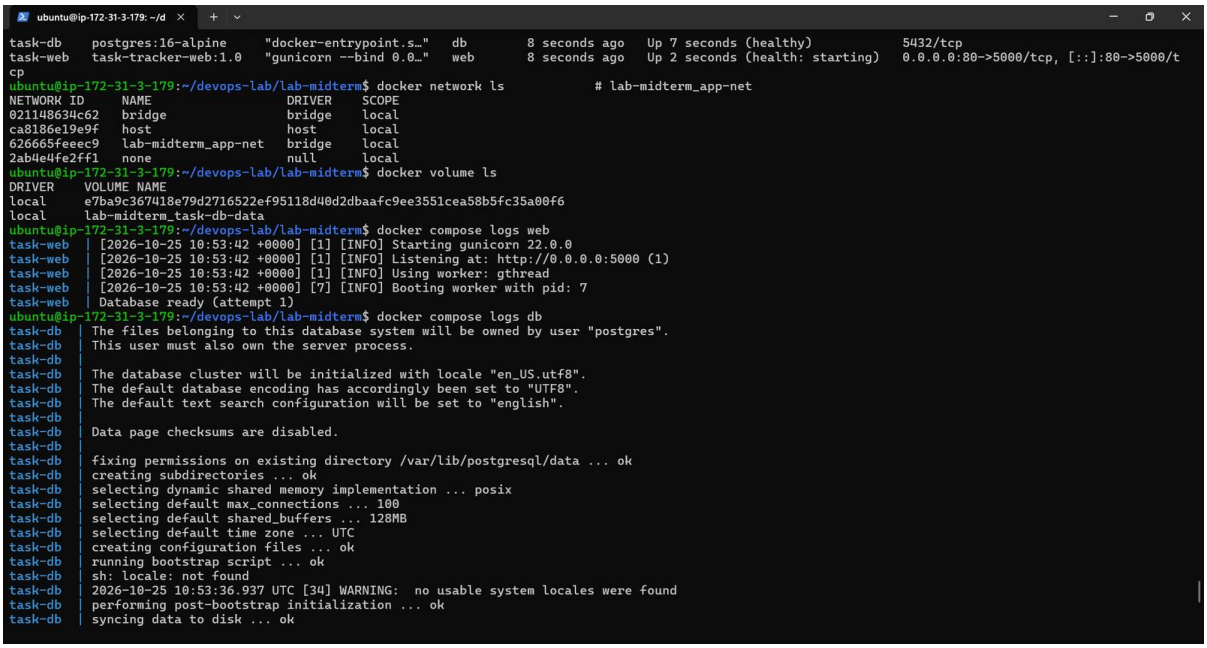


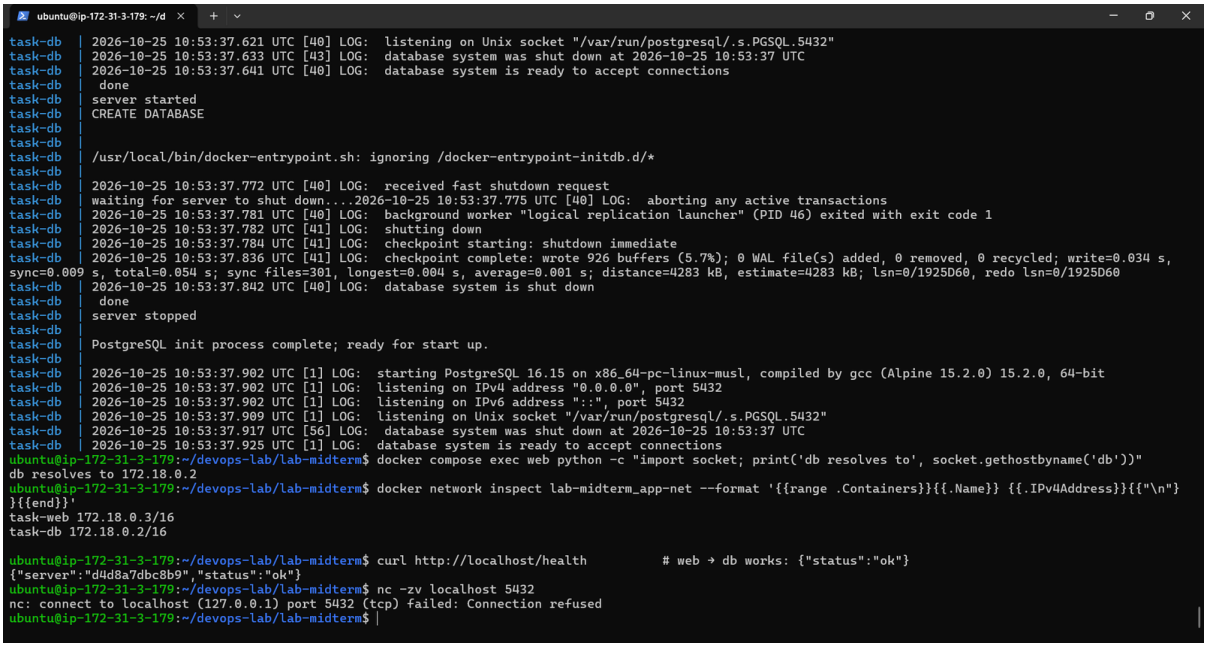
```
ubuntu@ip-172-31-3-179: ~/d
    retries: 10
  image: postgres:16-alpine
  networks:
    app-net: null
  restart: unless-stopped
  volumes:
    - type: volume
      source: task-db-data
      target: /var/lib/postgresql/data
      volume: {}
web:
  build:
    context: /home/ubuntu/devops-lab/lab-midterm
    dockerfile: Dockerfile
  container_name: task-web
  depends_on:
    db:
      condition: service_healthy
      required: true
  environment:
    DB_HOST: db
    POSTGRES_DB: tasks
    POSTGRES_PASSWORD
    POSTGRES_USER: taskuser
    SERVER_NAME: ""
  image: task-tracker-web:1.0
  networks:
    app-net: null
  ports:
    - mode: ingress
      target: 5000
      published: "80"
      protocol: tcp
  restart: unless-stopped
networks:
  app-net:
    name: lab-midterm_app-net
volumes:
  task-db-data:
    name: lab-midterm_task-db-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$
```

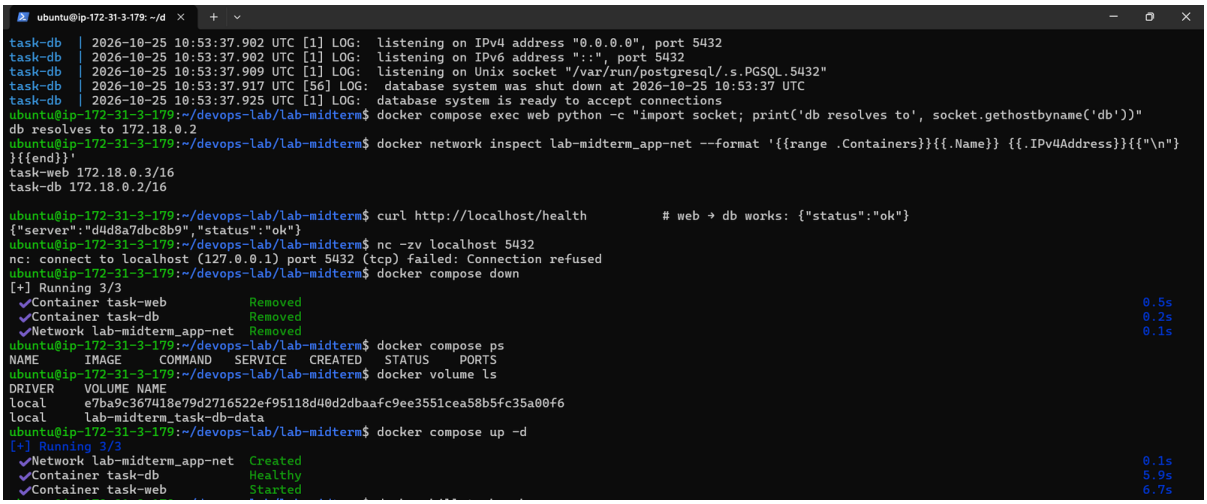


```
ubuntu@ip-172-31-3-179: ~/d
volumes:
  task-db-data:
    name: lab-midterm_task-db-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose up -d --build
WARN[0000] Docker Compose is configured to build using Bake, but buildx isn't installed
[+] Building 6.1s (12/12) FINISHED                                                                                                  docker:default
 => [web internal] load build definition from Dockerfile                                                                                      0.0s
 => => transferring dockerfile: 935B                                                                                                          0.0s
 => [web internal] load metadata for docker.io/library/python:3.12-slim                                                                       0.0s
 => [web internal] load .dockerignore                                                                                                         0.0s
 => => transferring context: 125B                                                                                                             0.0s
 => [web internal] load build context                                                                                                         0.1s
 => => transferring context: 6.75kB                                                                                                           0.0s
 => [web 1/6] FROM docker.io/library/python:3.12-slim@sha256:a0f57064cbdcd6285fa45f51237ab0b35ea8ebaed06a1f5973e89e19dfb86704                 0.1s
 => => resolve docker.io/library/python:3.12-slim@sha256:a0f57064cbdcd6285fa45f51237ab0b35ea8ebaed06a1f5973e89e19dfb86704                     0.0s
 => [web 2/6] WORKDIR /app                                                                                                                    0.0s
 => [web 3/6] COPY requirements.txt .                                                                                                         0.1s
 => [web 4/6] RUN pip install --no-cache-dir -r requirements.txt                                                                              3.7s
 => [web 5/6] COPY app/ .                                                                                                                     0.1s
 => [web 6/6] RUN useradd --create-home appuser                                                                                               0.3s
 => [web] exporting to image                                                                                                                  1.4s
 => => exporting layers                                                                                                                       1.0s
 => => exporting manifest sha256:beb94735a460a54bb52bb039a66aed56ec0533cfed046a2a9d8bd564b5b886d9                                             0.0s
 => => exporting config sha256:01c58f49c39de3cc2088d6feacc4a9116bcf3483c99152b63d1b3cfcaf5227eb                                               0.0s
 => => exporting attestation manifest sha256:b9816c33c941701210a7d2418f8bb84ea33ec1fc8642859925759b220df7738c                                 0.0s
 => => exporting manifest list sha256:0e0a0fd74ec76dbed725e7e3aca96305b2701218cf542252924cadde690d38a3                                        0.0s
 => => naming to docker.io/library/task-tracker-web:1.0                                                                                       0.0s
 => => unpacking to docker.io/library/task-tracker-web:1.0                                                                                    0.3s
 => [web] resolving provenance for metadata file                                                                                              0.0s
[+] Running 5/5
 ✔web                                Built                                                                                                   0.0s
 ✔Network lab-midterm_app-net        Created                                                                                                 0.1s
 ✔Volume lab-midterm_task-db-data    Created                                                                                                 0.0s
 ✔Container task-db                  Healthy                                                                                                 5.9s
 ✔Container task-web                 Started                                                                                                 6.0s
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose ps
NAME       IMAGE                    COMMAND                  SERVICE   CREATED         STATUS                            PORTS
task-db    postgres:16-alpine       "docker-entrypoint.s…"   db        8 seconds ago   Up 7 seconds (healthy)            5432/tcp
task-web   task-tracker-web:1.0     "gunicorn --bind 0.0…"   web       8 seconds ago   Up 2 seconds (health: starting)   0.0.0.0:80->5000/tcp, [::]:80->5000/t
cp
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$
```


Launch an instance | EC2 | ap-s
SecurityGroup | EC2 | ap-south
Student Task Tracker
Ask Gemini
Not secure
15.252.145.214
Domain Search | Blu...
Scholarships search...
Official Video Guide...
All Bookmarks
Student Task Tracker
Served by: d4d8a7dbc8b9
What do you need to do?
Medium
Add Task
3 task(s)
task 3
Medium
Done
Delete
task 2
Medium
Done
Delete
task 1
Medium
Done
Delete


ubuntu@ip-172-31-3-179: ~/d
task-db postgres:16-alpine "docker-entrypoint.s…" db 8 seconds ago Up 7 seconds (healthy) 5432/tcp
task-web task-tracker-web:1.0 "gunicorn --bind 0.0…" web 8 seconds ago Up 2 seconds (health: starting) 0.0.0.0:80->5000/tcp, [::]:80->5000/tcp
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network ls # lab-midterm_app-net
NETWORK ID NAME DRIVER SCOPE
021148634c62 bridge bridge local
ca8186e19e9f host host local
626665feeec9 lab-midterm_app-net bridge local
2ab4e4fe2ff1 none null local
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume ls
DRIVER VOLUME NAME
local e7ba9c367418e79d2716522ef95118d40d2dbaafc9ee3551cea58b5fc35a00f6
local lab-midterm_task-db-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose logs web
task-web | [2026-10-25 10:53:42 +0000] [1] [INFO] Starting gunicorn 22.0.0
task-web | [2026-10-25 10:53:42 +0000] [1] [INFO] Listening at: http://0.0.0.0:5000 (1)
task-web | [2026-10-25 10:53:42 +0000] [1] [INFO] Using worker: gthread
task-web | [2026-10-25 10:53:42 +0000] [7] [INFO] Booting worker with pid: 7
task-web | Database ready (attempt 1)
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose logs db
task-db | The files belonging to this database system will be owned by user "postgres".
task-db | This user must also own the server process.
task-db |
task-db | The database cluster will be initialized with locale "en_US.utf8".
task-db | The default database encoding has accordingly been set to "UTF8".
task-db | The default text search configuration will be set to "english".
task-db |
task-db | Data page checksums are disabled.
task-db |
task-db | fixing permissions on existing directory /var/lib/postgresql/data ... ok
task-db | creating subdirectories ... ok
task-db | selecting dynamic shared memory implementation ... posix
task-db | selecting default max_connections ... 100
task-db | selecting default shared_buffers ... 128MB
task-db | selecting default time zone ... UTC
task-db | creating configuration files ... ok
task-db | running bootstrap script ... ok
task-db | sh: locale: not found
task-db | 2026-10-25 10:53:36.937 UTC [34] WARNING: no usable system locales were found
task-db | performing post-bootstrap initialization ... ok
task-db | syncing data to disk ... ok


ubuntu@ip-172-31-3-179: ~/d
task-db | 2026-10-25 10:53:37.621 UTC [40] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
task-db | 2026-10-25 10:53:37.633 UTC [43] LOG: database system was shut down at 2026-10-25 10:53:37 UTC
task-db | 2026-10-25 10:53:37.641 UTC [40] LOG: database system is ready to accept connections
task-db | done
task-db | server started
task-db | CREATE DATABASE
task-db |
task-db |
task-db | /usr/local/bin/docker-entrypoint.sh: ignoring /docker-entrypoint-initdb.d/*
task-db |
task-db | 2026-10-25 10:53:37.772 UTC [40] LOG: received fast shutdown request
task-db | waiting for server to shut down....2026-10-25 10:53:37.775 UTC [40] LOG: aborting any active transactions
task-db | 2026-10-25 10:53:37.781 UTC [40] LOG: background worker "logical replication launcher" (PID 46) exited with exit code 1
task-db | 2026-10-25 10:53:37.782 UTC [41] LOG: shutting down
task-db | 2026-10-25 10:53:37.784 UTC [41] LOG: checkpoint starting: shutdown immediate
task-db | 2026-10-25 10:53:37.836 UTC [41] LOG: checkpoint complete: wrote 926 buffers (5.7%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.034 s, sync=0.009 s, total=0.054 s; sync files=301, longest=0.004 s, average=0.001 s; distance=4283 kB, estimate=4283 kB; lsn=0/1925D60, redo lsn=0/1925D60
task-db | 2026-10-25 10:53:37.842 UTC [40] LOG: database system is shut down
task-db | done
task-db | server stopped
task-db |
task-db | PostgreSQL init process complete; ready for start up.
task-db |
task-db | 2026-10-25 10:53:37.902 UTC [1] LOG: starting PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
task-db | 2026-10-25 10:53:37.902 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
task-db | 2026-10-25 10:53:37.902 UTC [1] LOG: listening on IPv6 address "::", port 5432
task-db | 2026-10-25 10:53:37.909 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
task-db | 2026-10-25 10:53:37.917 UTC [56] LOG: database system was shut down at 2026-10-25 10:53:37 UTC
task-db | 2026-10-25 10:53:37.925 UTC [1] LOG: database system is ready to accept connections
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose exec web python -c "import socket; print('db resolves to', socket.gethostbyname('db'))"
db resolves to 172.18.0.2
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network inspect lab-midterm_app-net --format '{{range .Containers}}{{.Name}} {{.IPv4Address}}{{"\n"}}{{end}}'
task-web 172.18.0.3/16
task-db 172.18.0.2/16
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ curl http://localhost/health # web → db works: {"status":"ok"}
{"server":"d4d8a7dbc8b9","status":"ok"}
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ nc -zv localhost 5432
nc: connect to localhost (127.0.0.1) port 5432 (tcp) failed: Connection refused
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$


ubuntu@ip-172-31-3-179: ~/d
task-db | 2026-10-25 10:53:37.902 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
task-db | 2026-10-25 10:53:37.902 UTC [1] LOG: listening on IPv6 address "::", port 5432
task-db | 2026-10-25 10:53:37.909 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
task-db | 2026-10-25 10:53:37.917 UTC [56] LOG: database system was shut down at 2026-10-25 10:53:37 UTC
task-db | 2026-10-25 10:53:37.925 UTC [1] LOG: database system is ready to accept connections
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose exec web python -c "import socket; print('db resolves to', socket.gethostbyname('db'))"
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ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker network inspect lab-midterm_app-net --format '{{range .Containers}}{{.Name}} {{.IPv4Address}}{{"\n"}}{{end}}'
task-web 172.18.0.3/16
task-db 172.18.0.2/16
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ curl http://localhost/health # web → db works: {"status":"ok"}
{"server":"d4d8a7dbc8b9","status":"ok"}
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ nc -zv localhost 5432
nc: connect to localhost (127.0.0.1) port 5432 (tcp) failed: Connection refused
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose down
[+] Running 3/3
✔Container task-web Removed 0.5s
✔Container task-db Removed 0.2s
✔Network lab-midterm_app-net Removed 0.1s
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker volume ls
DRIVER VOLUME NAME
local e7ba9c367418e79d2716522ef95118d40d2dbaafc9ee3551cea58b5fc35a00f6
local lab-midterm_task-db-data
ubuntu@ip-172-31-3-179:~/devops-lab/lab-midterm$ docker compose up -d
[+] Running 3/3
✔Network lab-midterm_app-net Created 0.1s
✔Container task-db Healthy 5.9s
✔Container task-web Started 6.7s

Task 6: Selenium Testing

```
PS D:\New folder> python tests\test_selenium.py http://task-tracker-alb-1135327115.ap-south-1.elb.amazonaws.com

DevTools listening on ws://127.0.0.1:52061/devtools/browser/11585488-d7c5-4573-9511-1179b3082456
PASS 1: page title is 'Student Task Tracker'
PASS 2: heading visible, served by i-08e26badae8565790
PASS 3: task 'Selenium task 1792929597' added and visible
PASS 4: task marked as done
PASS 5: task deleted

ALL 5 TESTS PASSED
PS D:\New folder>
```

```
PS D:\New folder> python tests\test_selenium.py http://task-tracker-alb-1135327115.ap-south-1.elb.amazonaws.com --headless

DevTools listening on ws://127.0.0.1:62154/devtools/browser/85ca4612-d7d1-4d03-99a3-5f289407025e
PASS 1: page title is 'Student Task Tracker'
PASS 2: heading visible, served by i-0b2586384003de901
PASS 3: task 'Selenium task 1792929643' added and visible
PASS 4: task marked as done
PASS 5: task deleted

ALL 5 TESTS PASSED
```